A Major Project Report on

"AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management"

SUBMITTED TO

THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

IN THE PARTIAL FULFILLMENT OF THE AWARD OF DEGREE

of

MASTER OF COMPUTER APPLICATION

(Academic Year 2025-26)

SUBMITTED BY

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UNDER THE GUIDANCE OF

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SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE

Academic Year 2025-26

BHUJBAL KNOWLEDGE CITY
Mumbai Educational Trust

Institute of Engineering

CERTIFICATE

This is to certify that the project report entitled

"AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management"

Ms. Yashika.J.Chawla EXAM SEAT NO : 1016

are bonafied students of this institute and work has been carried out by them under the guidance of Prof. Manjusha .V. Khond and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University for the award of Master degree of Computer Application (MCA).

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Date: January 01, 2026

To,
Yashika Jayprakash Chawla

Subject: Internship Offer Letter

Dear Yashika,

We are pleased to offer you the position of **Full Stack Web Development & Gen AI Intern** at **Engeniuspark Technologies Pvt. Ltd.** for the period **January 01, 2026 to June 06, 2026**. This internship will provide hands-on experience in full stack web development and help you develop essential technical and professional skills through real-world projects and collaborative learning.

During the course of your internship, your responsibilities will include:

- Assisting in automating and improving development and deployment processes
- Collaborating with development and operations teams to optimize system performance
- Troubleshooting and resolving infrastructure and deployment-related issues
- Maintaining documentation related to DevOps processes and best practices

Your working hours will be **8 hours per day**, and you are expected to comply with all company policies, rules, and professional ethics throughout the internship period. Upon successful completion of the internship, you will be awarded an **Internship Completion Certificate**.

Please confirm your acceptance of this offer by signing and returning a copy of this letter. We look forward to welcoming you to our team and supporting your professional growth.

Best regards,

HR & Administrator
Vishakha Patil
Authorized Signatory

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Internship Completion & Experience Certificate

Date: June 06, 2026

To Whom It May Concern,

This is to certify that Miss Yashika Chawla, a student of Master of Computer Applications (MCA) from MET Institute of Engineering, has successfully completed a 6-month internship in Full Stack Web Development and Generative AI at Engeniuspark Technologies Pvt. Ltd.. The internship was carried out from January 5, 2026 to June 6, 2026.

During her internship, Yashika demonstrated strong technical aptitude, problem-solving abilities and a proactive learning attitude. She was actively involved in the following key responsibilities:

Developed and maintained responsive web applications using frontend and backend technologies.

- Assisted in database management, API integration and debugging application issues.
- Collaborated using Git/GitHub for web application development and version control.
- Participated in testing, troubleshooting and deployment of web applications on cloud platforms.
- Supported ongoing projects involving Generative AI tools and automation features.

Projects Handled During Internship

- Independently developed the Digital Menu Card System, a QR-based restaurant management web application, where she handled the complete project lifecycle including frontend development, backend development, QR code management, API integration, database management, authentication, debugging and deployment support based on the provided project documentation and requirements.

- Worked on DecorBuddy, an AI-powered interior design web application, contributing to frontend development, responsive UI implementation, API integration and AI-based feature modules.

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- Contributed to the development of the Seeds IVF Website as part of a 5-member team, where she independently developed 2–3 web application pages including both frontend and backend functionality based on the provided premium template structure. The team also prepared the Scope of Work (SOW) documentation for the project.

- Worked on the Fortune Biomedicals Website, where she implemented product management functionalities including add, edit and delete operations along with backend integration and database connectivity.

- Assisted in developing Aarogya Sampada and other React.js-based web applications based on the provided project documentation and requirements, involving responsive frontend development, API integration and backend support functionalities.

During the internship, she worked with technologies including HTML, CSS, JavaScript, Tailwind CSS, React.js, Node.js, Express.js, MySQL, Git/GitHub and Generative AI tools.

Throughout the internship, Yashika showcased dedication, teamwork and an eagerness to learn, contributing positively to various ongoing projects. Her commitment, professionalism and ability to adapt to new technologies were highly commendable.

We appreciate her efforts and wish her continued success in all her future endeavors.

Sincerely,

For Engeniuspark Technologies Pvt. Ltd.
Nikita Kolhe
HR & Administrator

ACKNOWLEDGEMENT

I am sincerely thankful to all those who supported me during the development of this project titled **"AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management"**. Their guidance and encouragement helped me complete this work successfully.

I would like to sincerely thank my internal guide, **Prof. Manjusha V. Khond**, for her constant support and valuable suggestions throughout the project development. Her guidance helped me understand the requirements clearly and improve the implementation of the system.

I am grateful to the Head of the Department (MCA), **Prof. P. D. Jadhav**, and the Project Coordinator, **Prof. Sonali L. Vidhate**, for their support, guidance, and for providing the required resources to complete this project smoothly.

I would also like to thank all faculty members of the MCA Department for their help and guidance during this project. I am also thankful to my classmates, friends, and family members for their continuous encouragement and support.

Finally, I acknowledge everyone who supported me directly or indirectly in completing this project.

Ms. Yashika J. Chawla

Abstract

In today's digital age, restaurants are increasingly adopting smart technologies to improve customer convenience and service efficiency. This project presents a Digital Menu Application that uses QR code technology along with a mobile-friendly Progressive Web Application (PWA). Instead of traditional printed menus, customers can simply scan a table-specific QR code to access the menu directly on their smartphones. This allows them to view item details such as descriptions, prices, nutritional information, and dietary categories without requiring any account creation.

In addition to basic menu browsing, the system provides advanced features such as AI-based food recommendations, popular or trending items display, category and tag-based filtering, cart functionality, and real-time order tracking. The application also includes an administrative dashboard that supports menu updates, stock management, discount handling, and order monitoring, making restaurant operations more efficient.

By integrating modern web technologies, real-time communication features, and a user-friendly interface, this system aims to improve the overall dining experience and support the digital transformation of restaurant services.

Keywords— Digital Menu System, QR Code Ordering, Progressive Web Application (PWA), AI-Based Recommendation System, Restaurant Management System, Real-Time Order Tracking, Smart Dining Experience, Mobile-First Design, Food Personalization, Digital Transformation.

Contents

1 Introduction **1-7**

1.1 Company Profile 2-7

1.2 Company Working On 7

2 Literature Survey **8-10**

2.1 QR-Based Restaurant Ordering Systems and Digital Menus 8

2.2 AI-Based Recommendation Systems in Hospitality Applications 8-9

2.3 Progressive Web Applications in Modern Web Development 9

2.4 Real-Time Order Management and Customer Interaction Systems 9-10

2.5 Role-Based Restaurant Management Systems 10

2.6 Responsive User Interface and Visual-First Design 10

2.7 Summary 10

3 Problem Statement 11-13

3.1 Objectives and Scope 12-13

3.2 Summary 13

4 Analysis **14-17**

4.1 Project Plan 15

4.1.1 Project Plan for Semester IV 15

4.2 Requirement Analysis 16-17

4.2.1 Necessary Functions 16

4.2.2 Desirable Functions 16-17

4.3 Summary 17

5 Design 18-30

5.1 Software Requirement Specifications 18-22

5.1.1 Project Scope 18-19

5.1.2 Operating Environment 19-20

5.1.3 User Classes and Characteristics 20-21

5.1.4 Design and Implementation Constraints 21-22

5.2 Software Architechture 22-25

5.2.1 User Interface Layer 23

5.2.2 QR Access Layer 23

5.2.3 AI Recommendation Layer 23-24
5.2.4 Order Management Layer 24
5.2.5 Backend Processing Layer 24-25
5.2.6 Database layer 25
5.3 External Interface Requirement 25
5.3.1 User Interfaces 25
5.3.2 Hardware Interfaces 25
5.3.3 Communication Interfaces 25
5.4 Software System Attribute 25-26
5.5 Nonfunctional Requirement 26
5.6 Data Flow Diagram 26-29
5.7 Summary 30
6 Modeling 31-37
6.1 ER Diagram 31-32
6.2 Use Case Diagram 33
6.3 Class Diagram 34
6.4 Activity Diagram 35
6.5 Sequence Diagram 36
6.6 Summary 37
7 Implementation and Results 38-57
7.1 Implementation Details 38-54
7.1.1 User Interface Module 38-39
7.1.2 QR Access Module 40
7.1.3 AI Recommendation Module 41
7.1.4 Cart and Order Management Module 42
7.1.5 Favorites Module 43
7.1.6 Offers Module 44
7.1.7 Feedback Module 45
7.1.8 Backend Processing Module 46
7.1.9 Dashboard Management Module 47-49
7.1.9.1 Waiter Dashboard 47
7.1.9.2 Manager Dashboard 48
7.1.9.3 Owner Dashboard 49
7.1.10 Data Dictionary 50-54
7.1.11 Technology Stack 54

7.2 Results 54-57
7.2.1 Functional Results 55-56
7.2.2 Performance Results 56-57
7.2.3 Progress Evaluation 57
7.3 Summary 57

8 Testing 58-62
8.1 Formal Technical Review 58-59
8.2 Test Plan 59-61
8.2.1 Objectives of Testing 59
8.2.2 Testing Strategies 59-61
8.2.3 Sample Test Cases 61
8.3 Summary 62

9 Technical Specifications 63-65
9.1 Applications 63-64
9.1.1 Hardware Requirements 64
9.1.2 Software Requirements 64-65
9.2 Summary 65

10 Future Scope 66-67

11 Conclusion 68

Appendix **69-79**

A Glossary 69-72

B Achievements 73

C Plagiarism Report 74-78

D Team Information 79

Bibliography 80-81

List of Figures

1 System Architecture of AI-Assisted QR-Based Digital Menu Application 22
2 DFD Level 0 27
3 DFD Level 1 28
4 DFD Level 2 29
5 ER Diagram of AI-Assisted QR-Based Digital Menu Application 32
6 Use Case Diagram of AI-Assisted QR-Based Digital Menu Application 33
7 Class Diagram of AI-Assisted QR-Based Digital Menu Application 34
8 Activity Diagram of AI-Assisted QR-Based Digital Menu Application 35
9 Sequence Diagram of AI-Assisted QR-Based Digital Menu Application 36
10 QR Code-Based Restaurant Landing Page 39
11 Dining Ambience Selection Interface (Dine-In / Bar Menu) 39
12 QR-Based Menu Access System 40
13 AI Recommendation Assistant Interface 41
14 Cart and Real-Time Order Management Interface 42
15 Favorites Module Interface 43
16 Offers Module Interface 44
17 Customer Feedback and Review Interface 45
18 Complete System Workflow and Backend Processing Architecture of the QR-Based Restaurant Management System 46
19 Waiter Dashboard Interface 47
20 Manager Dashboard Interface 48
21 Manager Dashboard Interface 49
22 Users Table In Database 50
23 Categories Table In Database 50
24 Products Table In Database 51
25 Tags Table In Database 51
26 Product_Tags Table In Database 51
27 Tables Table In Database 52

28 QR_Codes Table In Database 52
29 Orders Table In Database 52
30 Order_Items Table In Database 53
31 Favorites Table In Database 53
32 Offers Table In Database 53
33 Feedbacks Table In Database 54
34 Waiter_Calls Table In Database 54
35 Dine-In-Menu Page 55
36 Bar Menu Page 56
37 Plagiarism Scan Report (1 of 5) – Overall Similarity Summary 74
38 Plagiarism Scan Report (2 of 5) 75
39 Plagiarism Scan Report (3 of 5) 76
40 Plagiarism Scan Report (4 of 5) 77
41 Plagiarism Scan Report (5 of 5) 78

List of Tables

1 Company Overview 3

2 Company Contact Information 6

3 Project Plan for Semester IV 15

4 Sample Test Cases for AI-Assisted QR-Based Digital Menu Application 61

5 Team Information 79

Chapter 1

Introduction

In today's digital era, technology plays a vital role in improving business operations and customer experiences across various industries. The restaurant and hospitality sector is also rapidly adopting digital solutions to provide faster, more efficient, and customer-friendly services. Traditional paper-based menu systems are gradually becoming outdated as they offer limited information and require customers to depend on restaurant staff for placing orders and obtaining details about food items. Traditional menu systems have several limitations, such as difficulty in updating menu content, lack of food images, and the inability to provide additional information like nutritional values, allergen details, dietary preferences, and special offers. During busy hours, customers often experience delays while waiting for menu explanations or order placement, which can affect overall customer satisfaction. Although some restaurants use QR-code menus, many of them only display static PDF documents and do not provide interactive features. To overcome these challenges, the **AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management** has been developed. The application provides a smart and interactive digital menu that customers can access by scanning a QR code using their smartphones. As a Progressive Web Application (PWA), the system can be accessed directly through a web browser without requiring installation of any mobile application. The platform allows customers to browse food items with detailed descriptions, prices, high-quality images, nutritional information, allergen details, and dietary categories. Advanced search and filtering options help users quickly find food items according to their preferences, such as Vegetarian, Vegan, Jain, Gluten-Free, or Spicy dishes. This improves the overall dining experience and helps customers make informed decisions. A major feature of the system is the integration of Artificial Intelligence (AI) for personalized food recommendations. The AI assistant analyzes customer preferences and interactions to suggest suitable dishes, making menu selection easier and more engaging. Customers can

also add items to their cart, customize food options, place orders digitally, and track order status in real time. The system follows a role-based architecture consisting of Customers, Waiters, Managers, and Restaurant Administrators. Each user has specific functionalities for managing orders, menu items, customer requests, and restaurant operations. The application also includes features such as real-time order tracking and a "Call Waiter" facility to improve communication between customers and restaurant staff. Overall, the AI-Assisted QR-Based Digital Menu Application provides a modern and efficient solution for restaurant management. By combining QR technology, Artificial Intelligence, and digital ordering capabilities, the system enhances customer satisfaction, improves operational efficiency, and supports the digital transformation of the restaurant industry.

1.1 Company Profile

Company Name : Engeniuspark Technologies Pvt. Ltd. (ESTPL)

Introduction to the Company

Engeniuspark Technologies Pvt. Ltd. (ESTPL) is a leading Information Technology company located in Nashik, Maharashtra, India. Established in 2014, the company provides innovative and affordable technology solutions to businesses across various industries. ESTPL specializes in Software Development, Web Application Development, Mobile Application Development, Artificial Intelligence (AI), Internet of Things (IoT), Digital Marketing, and Smart Technology Solutions.

The company is committed to delivering high-quality services by adopting the latest technologies and industry best practices. With a customer-centric approach, ESTPL focuses on understanding client requirements and developing customized solutions that enhance business efficiency and productivity. The organization emphasizes innovation, teamwork, professionalism, and continuous improvement to ensure excellence in every project.

Over the years, ESTPL has successfully delivered numerous projects for both national and international clients. Its strong technical expertise, timely project delivery, and dedication to customer satisfaction have helped the company establish a strong presence in India as well as in international markets such as the USA and UAE. Through its commitment

to technological advancement and quality service, ESTPL continues to contribute to the digital transformation of businesses across various sectors.

Company Overview

Particular	**Details**
Company Name	Engeniuspark Technologies Pvt. Ltd. (ESTPL)
Industry	Information Technology and Software Services
Founded	2014
Headquarters	Nashik, Maharashtra, India
Company Type	Private Limited Company
Website	www.estplnow.com
Core Domains	Software Development, AI, IoT, Mobile Applications, Digital Marketing
Presence	India, USA, UAE
Client Base	National and International Clients

Table 1 : Company Overview

History and Background

Vision

To become a globally recognized technology organization by delivering innovative, reliable, and sustainable digital solutions that empower businesses and contribute to technological advancement.

Mission

- To provide high-quality software and technology solutions.
- To leverage emerging technologies for business growth.
- To maintain excellence through innovation and continuous improvement.

- To build long-term relationships with clients through trust and professionalism.
- To deliver cost-effective and customer-centric services.

Core Values

- Innovation
- Integrity
- Customer Satisfaction
- Excellence
- Teamwork
- Continuous Learning
- Accountability
- Professionalism

Services Offered

- Software Development
- Web Development
- Mobile Application Development
- Artificial Intelligence Solutions
- Internet of Things (IoT)
- Digital Marketing
- Aviation Solutions
- Testing and Quality Assurance

Organizational Structure

- Chief Executive Officer (CEO)
- Directors
- Project Managers
- Team Leaders
- Software Developers
- UI/UX Designers
- Quality Assurance Engineers
- Database Administrators
- Digital Marketing Team
- Technical Support Team
- Human Resource Department

Technologies Used

Frontend Technologies: HTML5, CSS3, JavaScript, React.js
Backend Technologies: Node.js, Express.js, Java, .NET Technologies
Database Technologies: MySQL, MongoDB
Development Tools: Visual Studio Code, Git, GitHub, Postman
Emerging Technologies: Artificial Intelligence, Internet of Things (IoT), Cloud Computing

Why Choose Engeniuspark Technologies Pvt. Ltd.

- Experienced and highly skilled professionals.
- Customer-centric development approach.
- Timely project delivery.

- Innovative and scalable solutions.
- Strong technical support.
- Global presence and industry expertise.
- Cost-effective services.
- Quality-driven development process.

Strengths of the Company

- Experienced technical professionals with strong domain knowledge.
- Focus on innovation and adoption of latest technologies.
- High-quality software development with efficient project management.
- Strong customer support and global client reach.

Contact Information

Particular	**Details**
Company Name	Engeniuspark Technologies Pvt. Ltd.
Address	Office No. 403, Shree Kalika, Opp. Kalika Mata Mandir, Above Haldiram & Zudio, Shingada Talav, Mumbai Naka, Renuka Nagar, Nashik, Maharashtra – 422011
Contact Number	+91 90117 55405
Website	www.estplnow.com
Global Presence	India, USA, UAE

Table 2 : Company Contact Information

Internship Exposure

In the course of the internship at Engeniuspark Technologies Pvt. Ltd., practical exposure was gained in software development methodologies, project management practices, web technologies, database management systems, and industrial development tools. The internship helped gain insights into actual software development environments and the professional work atmosphere.

Conclusion

Engeniuspark Technologies Pvt. Ltd. has been able to position itself as an emerging and innovative technology firm owing to the provision of top-notch digital services in diverse sectors. The company is committed to innovation, customer satisfaction, and technological superiority, which has enabled it to make significant contributions to the digital revolution.

1.2 Company Working On

The major working activities of Engeniuspark Technologies Pvt. Ltd. (ESTPL) are as follows:

1. Understanding and analyzing client requirements for software and technology solutions.
2. Designing and developing web and mobile applications using modern technologies.
3. Implementing customized software solutions including AI and IoT-based systems.
4. Managing databases and ensuring efficient data storage and retrieval.
5. Performing software testing to ensure quality, performance, and reliability.
6. Deploying applications and providing maintenance and technical support to clients.
7. Following Agile and Scrum methodologies for efficient project management and timely delivery of projects.
8. Researching and adopting emerging technologies to improve software quality and deliver innovative solutions.

Chapter 2

Literature Survey

In this chapter, different research works, technologies, and existing digital solutions related to AI-powered restaurant systems, QR-based ordering platforms, recommendation systems, Progressive Web Applications (PWA), and intelligent customer interaction technologies are studied. The literature survey helps in understanding the recent progress in digital restaurant solutions and highlights the importance of integrating artificial intelligence, responsive web design, and real-time communication features into modern hospitality systems. The reviewed studies form the foundation for developing the AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management.

2.1 QR-Based Restaurant Ordering Systems and Digital Menus

Many restaurants and hospitality businesses are adopting QR-code-based digital menu systems to reduce physical contact, improve hygiene, and make menu access more convenient for customers. In these systems, customers scan a QR code placed on their table and instantly access the digital menu on their mobile devices without installing any application. Research shows that QR-based menus help reduce printing costs, improve accessibility, and allow real-time menu updates.

However, most existing systems provide only basic menu listings or static PDF menus. They lack advanced features such as intelligent recommendations, real-time order tracking, and interactive customer engagement. Due to these limitations, the user experience remains simple and less personalized. The proposed system improves these limitations by integrating AI-based recommendations, smart filtering, and real-time order management into a single platform.

2.2 AI-Based Recommendation Systems in Hospitality Applications

Artificial Intelligence-based recommendation systems are widely used in modern digital platforms as they enhance user experience through personalized suggestions. These

systems are commonly used in e-commerce platforms, streaming services, and food delivery applications to recommend relevant items based on user preferences and behavior.
In restaurant applications, AI-based recommendation systems analyze customer preferences, dietary needs, browsing activity, and past orders to suggest suitable food items. Studies indicate that personalized recommendations improve customer satisfaction, simplify decision-making, and increase order conversion rates. However, many existing restaurant systems still rely on manual browsing and lack intelligent recommendation capabilities. The proposed system introduces an AI Recommendation Assistant that suggests food items based on mood, taste preferences, dietary requirements, and user interaction patterns.

2.3 Progressive Web Applications in Modern Web Development

Progressive Web Applications (PWAs) are a modern alternative to traditional mobile applications. They provide app-like experiences directly through web browsers without requiring installation from app stores. PWAs offer several benefits such as fast loading speed, responsive design, offline support, cross-platform compatibility, and reduced development effort.
These advantages make PWAs suitable for restaurant applications where users need quick and seamless access. Existing restaurant systems built using PWA architecture show improved accessibility and better user engagement. Based on these benefits, the proposed system uses a mobile-first PWA approach to ensure smooth performance across smartphones, tablets, and desktop devices with a responsive interface.

2.4 Real-Time Order Management and Customer Interaction Systems

Real-time communication systems are essential in improving restaurant efficiency and customer satisfaction. Modern restaurant platforms use real-time technologies to manage orders, track table activity, update order status, and enable communication between customers and restaurant staff.
Real-time order tracking improves transparency and reduces waiting frustration. Additionally, waiter notification systems and live dashboards help staff respond quickly to customer requests and improve workflow efficiency. However, many existing systems still depend on manual communication methods, leading to delays and operational issues. The proposed system includes real-time order updates, waiter call functionality, and live

order management for improved dining experience.

2.5 Role-Based Restaurant Management Systems

Role-based management systems are commonly used in enterprise and hospitality applications to improve security and operational control. In restaurant environments, different users such as customers, waiters, managers, and administrators perform different roles. Research shows that role-based access control improves efficiency by restricting system features based on user roles and reducing operational confusion. Systems with separate dashboards for different roles provide better monitoring and workflow management. The proposed system includes a structured role-based design consisting of Customers, Waiters, Managers, and Restaurant Owners/Admins to ensure smooth restaurant operations.

2.6 Responsive User Interface and Visual-First Design

User Interface (UI) and User Experience (UX) design play an important role in the success of digital restaurant applications. Studies show that attractive food presentation increases customer engagement and influences ordering decisions. Users prefer clean, responsive, and interactive interfaces with smooth navigation and visually appealing layouts.
Modern applications using card-based layouts, high-quality food images, filtering options, and personalized interactions provide better user experience and higher retention rates. However, many existing systems still use simple and cluttered interfaces that reduce engagement. The proposed system focuses on a visual-first design approach with responsive layouts, smart navigation, food cards, lazy loading of images, and modern UI components optimized for mobile devices.

2.7 Summary

This chapter reviewed different research studies and technological advancements related to QR-based restaurant systems, AI recommendation engines, Progressive Web Applications, real-time communication systems, role-based management systems, and modern UI/UX design. The literature survey shows that existing systems provide basic digital solutions but lack advanced personalization, real-time interaction, and integrated management features. The proposed AI-Assisted QR-Based Digital Menu Application addresses these limitations by combining QR-based access, AI-based recommendations, real-time order tracking, responsive PWA architecture, and role-based management into a unified system for modern restaurants.

Chapter 3

Problem Statement

Modern restaurants and hospitality businesses often encounter difficulties in managing customer interactions, menu accessibility, order processing, and overall operational efficiency when relying on traditional systems. Conventional paper menus and manual ordering processes are not only time-consuming but also challenging to update and maintain. These methods fail to support interactive and personalized dining experiences expected by today's customers.

As a result, customers frequently face delays in receiving menu assistance, confirming orders, or getting waiter support, particularly during busy hours. Furthermore, traditional restaurant systems do not provide real-time communication, intelligent food recommendations, advanced filtering options, or smooth digital interaction features, which are essential in modern smart restaurant environments.

Although many restaurants have introduced QR-based menu systems, most existing solutions are limited to static PDF menus or basic digital catalogues that do not support intelligent interaction, personalized recommendations, or real-time operational management. These systems generally fail to provide immersive user experiences, dynamic menu updates, dietary-based filtering, or customer engagement features. Furthermore, restaurants struggle with operational challenges such as inefficient order coordination, delayed communication between customers and staff, manual stock handling, and the absence of centralized management systems for promotions, analytics, and live order tracking.

The absence of AI-assisted recommendation systems also limits customer personalization and food discovery. Customers are often unable to identify suitable dishes according to their dietary preferences, taste choices, or nutritional requirements. Existing systems additionally provide limited accessibility across devices and frequently require users to install mobile applications, creating unnecessary friction during customer interaction.

3.1 Objectives and Scope

Objectives

- To develop a QR-based digital menu platform with a responsive and visually immersive user interface that supports seamless restaurant browsing across smartphones, tablets, and desktop devices.

- To implement a Progressive Web Application (PWA) architecture that allows customers to access the system directly through web browsers without downloading any mobile application.

- To build an AI-Assisted Recommendation System capable of suggesting personalized food items according to customer preferences, dietary requirements, browsing behaviour, and mood-based interactions.

- To develop advanced menu browsing features including category-based navigation, smart search functionality, dietary filters, allergen information, and dynamic menu customization.

- To implement a real-time cart and order management system that enables customers to place orders, customize food variants, and monitor live order status updates efficiently.

- To integrate a "Call Waiter" communication feature for improving coordination and reducing service delays between customers and restaurant staff.

- To design a role-based management architecture consisting of Customers, Waiters, Managers, and Masters (Restaurant Admins) for efficient operational control and restaurant workflow management.

- To develop administrative dashboards for menu management, discount handling, stock updates, analytics monitoring, user management, and restaurant configuration.

- To ensure system security using role-based authentication, secure session handling, and protected communication mechanisms.

- To improve customer engagement, operational efficiency, and overall dining experience through AI-assisted interaction and modern responsive web technologies.

Scope

The AI-Assisted QR-Based Digital Menu Application focuses on providing an intelligent and interactive restaurant management solution for modern hospitality environments. The system is designed for restaurants, cafés, food courts, hotels, and smart dining establishments that require digital menu accessibility, AI-assisted customer interaction, and real-time order management. The platform supports seamless customer access through QR-code scanning and enables restaurants to manage menu content, promotions, stock availability, customer interaction, and operational workflows efficiently.

The project scope additionally includes responsive Progressive Web Application support, real-time communication mechanisms, AI-powered recommendation features, role-based dashboards, and scalable restaurant management functionalities. Future enhancements of the system may include multilingual support, online payment gateway integration, loyalty reward systems, cloud-native deployment, customer behaviour analytics, and third-party delivery platform integration for large-scale restaurant ecosystems.

3.2 Summary

In this chapter, the major problems associated with traditional restaurant menu systems and existing QR-based solutions were discussed. The limitations related to manual ordering, lack of personalization, inefficient communication, and absence of intelligent interaction highlighted the need for a smarter digital restaurant platform. The objectives and scope of the AI-Assisted QR-Based Digital Menu Application define how the system utilizes QR accessibility, AI-powered recommendations, real-time order tracking, responsive PWA architecture, and role-based management modules to improve restaurant operations and deliver an enhanced dining experience for customers and restaurant staff.

Chapter 4

Analysis

This chapter describes the project planning and requirement analysis carried out during the development of the AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management. The system was developed using an iterative and modular software development approach that allowed the gradual implementation of customer interaction modules, AI-assisted recommendation systems, real-time order management, and administrative dashboards. The project architecture follows a structured client-server model that separates the frontend user interface, backend business logic, real-time communication services, and database management components for better scalability and maintainability.

The frontend application was designed using modern web technologies focused on responsive and mobile-first user experience principles. The customer-facing interface emphasizes smooth navigation, visually immersive food presentation, and QR-based accessibility. The backend services were planned to support menu management, role-based authentication, order handling, recommendation logic, and real-time communication between customers and restaurant staff. The requirement analysis phase involved identifying operational problems faced by restaurants, studying existing digital menu solutions, and understanding customer expectations related to convenience, personalization, and real-time interaction.

The development process was divided into multiple phases to ensure systematic implementation of the project modules. Initial phases focused on problem identification, literature survey, technology stack selection, requirement analysis, and user interface planning. Later phases concentrated on implementing QR-based menu access, AI-assisted recommendations, role-based dashboards, real-time order management, and responsive Progressive Web Application functionality. Feedback from project guides, development team members, and representative end users was considered throughout the planning and analysis process to improve usability and operational efficiency.

4.1 Project Plan

4.1.1 Project Plan for Semester IV

The development of the AI-Assisted QR-Based Digital Menu Application was organized into multiple stages during Semester I. The first semester mainly focused on identifying restaurant management challenges, studying existing digital menu systems, selecting appropriate technologies, designing the system architecture, and creating the initial user interface and operational modules.

Activity	**Start Date**	**End Date**	**Group Members**
Client Requirement Discussion and Project Selection	15-04-2026	17-04-2026	Team
Problem Statement and Objective Definition	18-04-2026	22-04-2026	Team
Literature Survey on Digital Menu and AI Systems	23-04-2026	30-04-2026	Team
Study of QR Systems, PWAs, and Recommendation Engines	01-05-2026	07-05-2026	Individual
Technology Stack Selection and Requirement Analysis	08-05-2026	14-05-2026	Individual
System Architecture and Database Design	15-05-2026	20-05-2026	Individual
UML Diagrams (Use Case, Activity, Sequence, Class Diagram)	21-05-2026	25-05-2026	Individual
Initial UI Design and QR Menu Prototype Development	26-05-2026	30-05-2026	Individual

Table 3 : Project Plan for Semester IV

4.2 Requirement Analysis

4.2.1 Necessary Functions

- QR-Based Digital Menu Accessibility.
- Responsive Progressive Web Application (PWA) Interface.
- AI-Assisted Food Recommendation System.
- Advanced Menu Search and Filtering Features.
- Real-Time Cart and Order Management System.
- Live Order Status Tracking.
- "Call Waiter" Communication Feature.
- Role-Based Authentication and Access Control.
- Menu and Inventory Management Dashboard.
- Discount, Promotion, and Offer Management.
- Restaurant Analytics and Monitoring Dashboard.
- Secure Session Handling and Data Protection.

4.2.2 Desirable Functions

- Multilingual Support for Regional Languages.
- Online Payment Gateway Integration.

- Loyalty Reward and Membership System.
- Customer Review and Feedback System.
- AI-Based Customer Behaviour Analytics.
- Third-Party Food Delivery Platform Integration.
- Cloud-Based Auto-Scaling Deployment Support.

4.3 Summary

In this chapter, the project planning and requirement analysis for the AI-Assisted QR-Based Digital Menu Application were discussed. The analysis identified the major functional and operational requirements necessary for building an intelligent restaurant management platform. Important system features such as QR-based accessibility, AI-assisted recommendations, real-time order tracking, role-based dashboards, responsive PWA architecture, and operational management modules were analyzed in detail. These requirements provide a strong foundation for developing a scalable, interactive, and efficient smart restaurant management system capable of improving both customer experience and restaurant operations.

Chapter 5

Design

This chapter describes the Software Requirement Specification (SRS) and overall system design of the AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management. It explains the system architecture, external interface requirements, operational constraints, and the overall data flow structure of the proposed system. The chapter additionally describes how different modules such as QR-based access, AI-assisted recommendations, order management, role-based dashboards, and real-time communication components interact to provide a seamless and intelligent restaurant management experience.

The system was designed using a modular and scalable architecture that separates frontend presentation, backend processing, database management, and real-time communication services into independent layers. This separation improves maintainability, scalability, and system performance while allowing individual modules to be updated or expanded without affecting the entire platform. The project follows modern web application development principles and uses responsive Progressive Web Application (PWA) architecture to ensure accessibility across multiple devices and operating environments.

5.1 Software Requirement Specifications

The Software Requirement Specification (SRS) defines the functional scope, operating environment, user characteristics, system limitations, and technical requirements of the AI-Assisted QR-Based Digital Menu Application. The SRS acts as the foundation for system implementation and helps ensure that all operational and user requirements are properly addressed during development.

5.1.1 Project Scope

The AI-Assisted QR-Based Digital Menu Application addresses the limitations of traditional restaurant menu systems and basic QR-based digital solutions by providing a smart,

interactive, and AI-assisted restaurant management platform. Modern customers expect fast service, visually engaging interfaces, personalized recommendations, and seamless ordering experiences through their smartphones. Existing restaurant systems often fail to provide real-time interaction, intelligent recommendations, and operational efficiency required in modern hospitality environments.

This project focuses on developing a web-based restaurant management platform that combines QR-based accessibility, AI-powered food recommendations, responsive Progressive Web Application architecture, and real-time order management into a unified digital solution. The system allows customers to scan QR codes placed at restaurant tables and instantly access an interactive digital menu without downloading any application.

The major scope of the project includes:

- Providing customers with an interactive QR-based digital menu platform accessible through smartphones and web browsers.
- Implementing AI-assisted recommendation features for personalized food suggestions based on customer preferences and dietary requirements.
- Enabling real-time order management, live order tracking, and customer-waiter communication.
- Providing role-based dashboards for Customers, Waiters, Managers, and Restaurant Admins.
- Supporting menu management, stock updates, promotions, analytics, and operational monitoring through administrative interfaces.

5.1.2 Operating Environment

The AI-Assisted QR-Based Digital Menu Application is designed to operate on modern web browsers across desktop and mobile devices without requiring dedicated application installation.

Software Environment:

- Frontend: React.js with Tailwind CSS and React Router
- Backend: Node.js with Express.js
- Database: MySql Workbench
- Real-Time Communication: Socket.IO / WebSocket
- AI Recommendation Engine: Machine Learning Recommendation Logic
- Deployment: Vercel / Cloud Hosting Platform

Hardware Environment:

- Minimum 4 GB RAM (8 GB Recommended)
- Smartphone, Tablet, or Desktop Device
- Stable Internet Connection
- QR Scanner Enabled Smartphone Camera

5.1.3 User Classes and Characteristics

The AI-Assisted QR-Based Digital Menu Application supports multiple user classes with different levels of system access and operational responsibilities.

- Customers: Restaurant visitors who scan QR codes, browse menus, receive AI-assisted recommendations, place orders, and track order status.
- Waiters: Restaurant staff members responsible for monitoring orders, responding to waiter calls, and updating order statuses.

- Managers: Authorized personnel responsible for promotions, discounts, stock monitoring, analytics, and restaurant operations.
- Masters / Restaurant Admins: Users with complete administrative access responsible for menu management, user management, system settings, and operational configuration.
- Developers: Technical personnel responsible for backend maintenance, deployment, database optimization, and feature enhancements.

5.1.4 Design and Implementation Constraints

The development of the AI-Assisted QR-Based Digital Menu Application involves several technical and operational constraints.

- Internet Dependency: The system requires stable internet connectivity for real-time communication and database synchronization.
- Real-Time Performance: Order updates and communication must occur with minimal delay to ensure smooth restaurant operations.
- Device Compatibility: The platform must function efficiently across different screen sizes and operating systems.
- Security Requirements: Authentication, session handling, and role-based access control must be implemented securely.
- Recommendation Accuracy: AI recommendation quality depends on customer interaction data and menu metadata.

The following are the major advantages of the proposed design implementation:

- Improved Customer Experience through interactive digital menus and AI-assisted recommendations.

- Real-Time Restaurant Operations Management.
- Reduced Waiter Workload and Faster Service.
- Scalable and Modular System Architecture.
- Cross-Platform Accessibility through Progressive Web Application support.

5.2 System Architecture

The AI-Assisted QR-Based Digital Menu Application follows a layered and modular architecture consisting of multiple interconnected system components. The architecture is divided into major layers including User Interface Layer, QR Access Layer, AI Recommendation Layer, Order Management Layer, Backend Processing Layer, and Database Layer. Each layer performs a specific role and communicates with other modules through secure APIs and real-time communication channels.

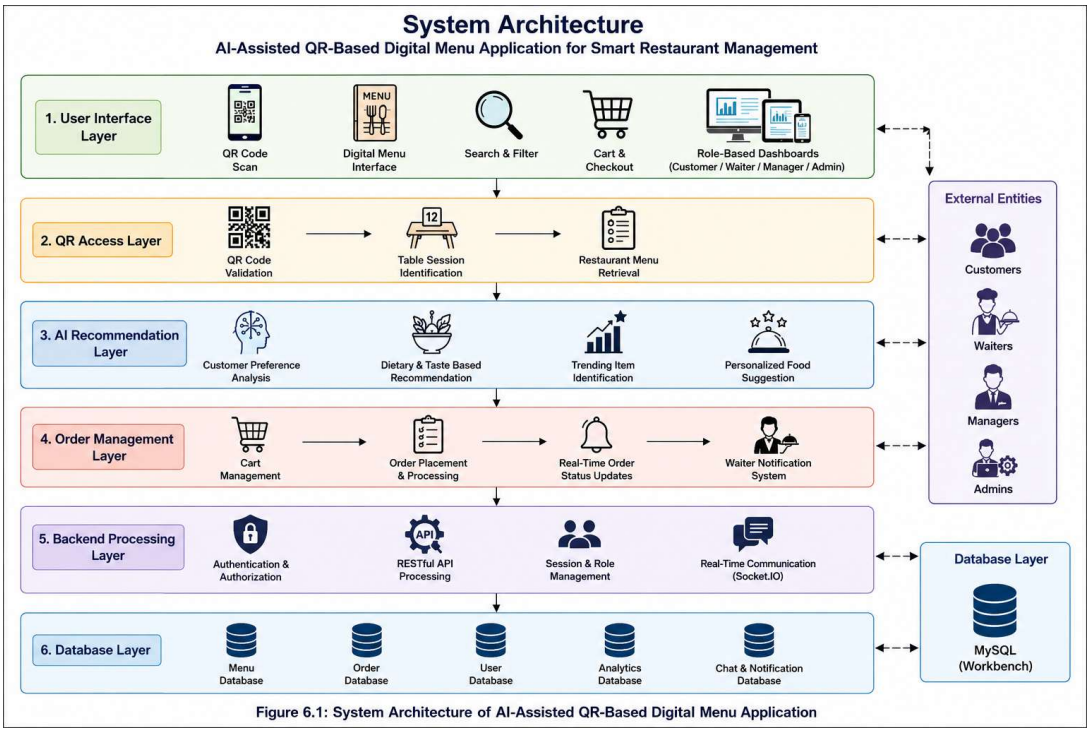


Figure 1: System Architecture of AI-Assisted QR-Based Digital Menu Application

5.2.1 User Interface Layer

This layer handles all customer and administrative interaction through the responsive React.js web application.

Components:

- Customer Digital Menu Interface
- QR-Based Access System
- Search and Filtering Interface
- Cart and Checkout Interface
- Role-Based Administrative Dashboards

5.2.2 QR Access Layer

This layer manages QR-code scanning and customer session initialization. Customers scan table QR codes to access restaurant-specific menu information directly through the browser.

Functions:

- QR Code Validation
- Table Session Identification
- Restaurant Menu Retrieval

5.2.3 AI Recommendation Layer

This layer handles intelligent food recommendation and personalization features based on customer preferences and interaction behaviour.

Functions:

- Customer Preference Analysis
- Dietary and Taste-Based Recommendation
- Trending Item Identification
- Personalized Food Suggestion Generation

5.2.4 Order Management Layer

This layer manages customer orders, waiter communication, and live order tracking functionalities.

Functions:

- Cart Management
- Order Placement and Processing
- Real-Time Order Status Updates
- Waiter Notification System

5.2.5 Backend Processing Layer

This layer manages business logic, authentication, API handling, session management, and communication between frontend and database services.

Functions:

- Authentication and Authorization
- RESTful API Processing
- Session and Role Management

– Real-Time Communication Handling

5.2.6 Database Layer

This layer stores all restaurant-related information including menu details, orders, customer interactions, user accounts, analytics, and operational records.

5.3 External Interface Requirement

5.3.1 User Interfaces

– Customer Interface: Allows customers to browse menu items, apply filters, receive recommendations, place orders, and track order status.

– Admin Dashboard: Allows restaurant staff to manage menu items, promotions, stock availability, analytics, and operational settings.

5.3.2 Hardware Interfaces

– Smartphone Camera for QR scanning.

– Desktop or Mobile Device with modern browser support.

– Internet Connectivity for real-time communication.

5.3.3 Communication Interfaces

The system communicates using HTTPS protocols, RESTful APIs, and real-time WebSocket communication for secure and efficient frontend-backend interaction.

5.4 Software System Attribute

– Reliability: Ensures smooth order handling and stable customer interaction.

– Availability: Cloud-based deployment ensures continuous accessibility.

- Maintainability: Modular architecture allows easier updates and maintenance.
- Portability: The application works across smartphones, tablets, and desktop browsers.

5.5 Nonfunctional Requirement

- Performance: The system should process customer interactions and order updates efficiently with minimal delay.
- Usability: Interfaces should remain intuitive and easy to use without training.
- Scalability: The architecture should support increasing restaurant traffic and operational expansion.
- Security: User sessions and restaurant data must remain secure and protected.
- Responsiveness: The application should adapt efficiently across different devices and screen resolutions.

5.6 Data Flow Diagram

The Data Flow Diagram (DFD) of the AI-Assisted QR-Based Digital Menu Application illustrates how customer requests, menu information, AI recommendation logic, and order management data move through the system. Customers scan QR codes to access menu data, interact with menu items, and place orders. The backend processes these requests, communicates with the database, and generates outputs such as recommendations, order confirmations, and operational analytics. The DFD additionally represents interactions between customers, waiters, managers, admins, and system databases.

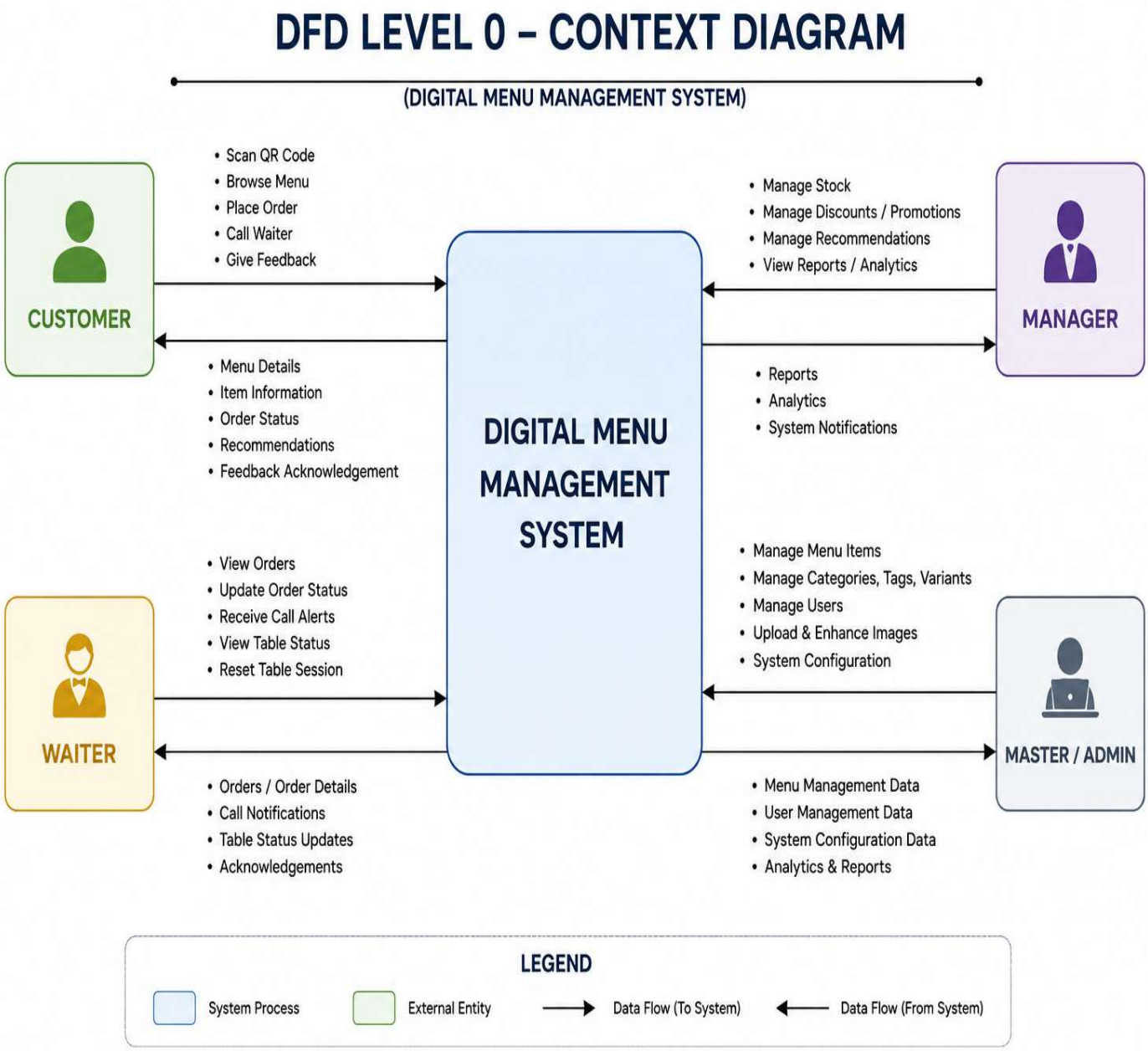


Figure 2: DFD Level 0

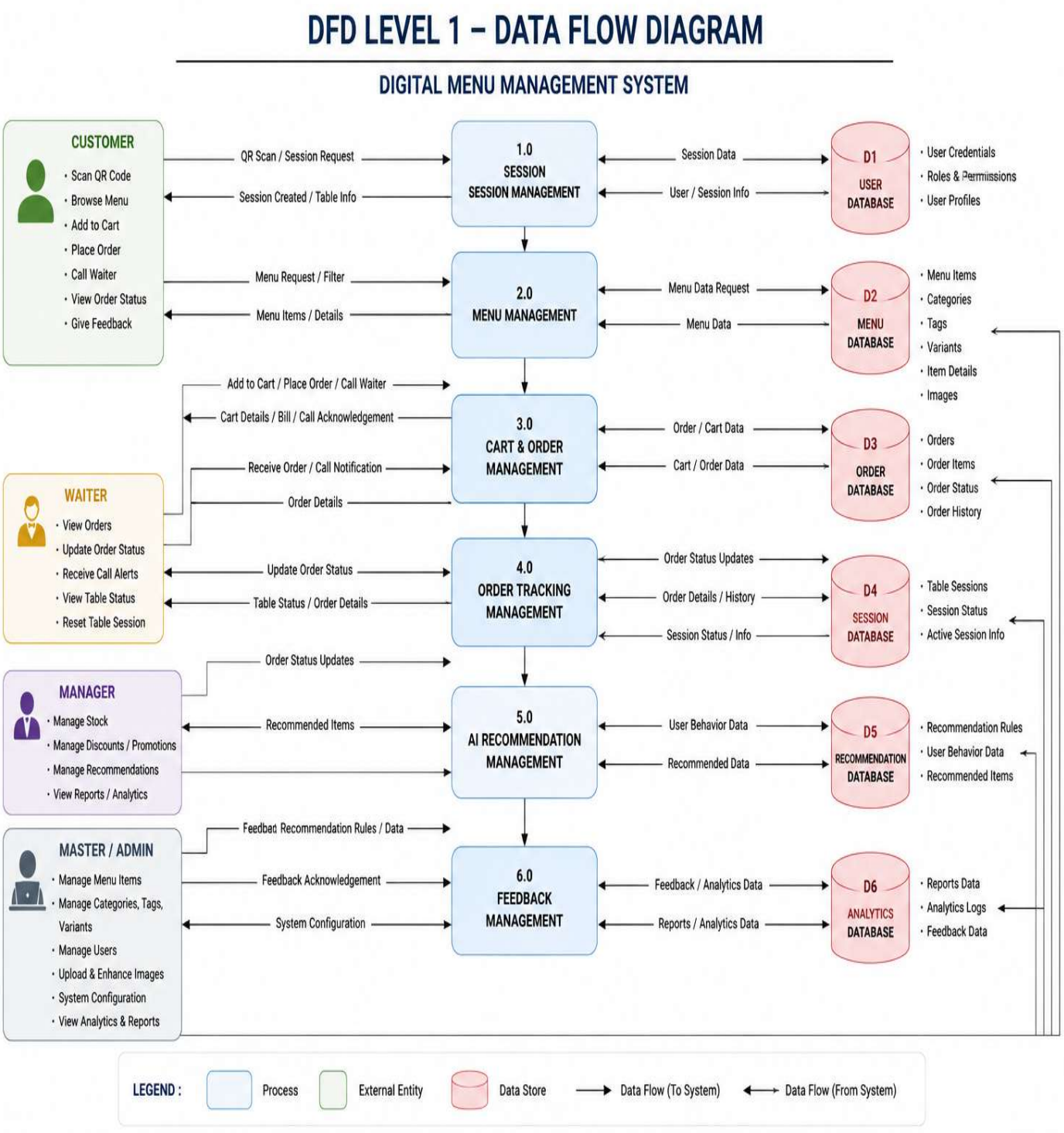


Figure 3: DFD Level 1

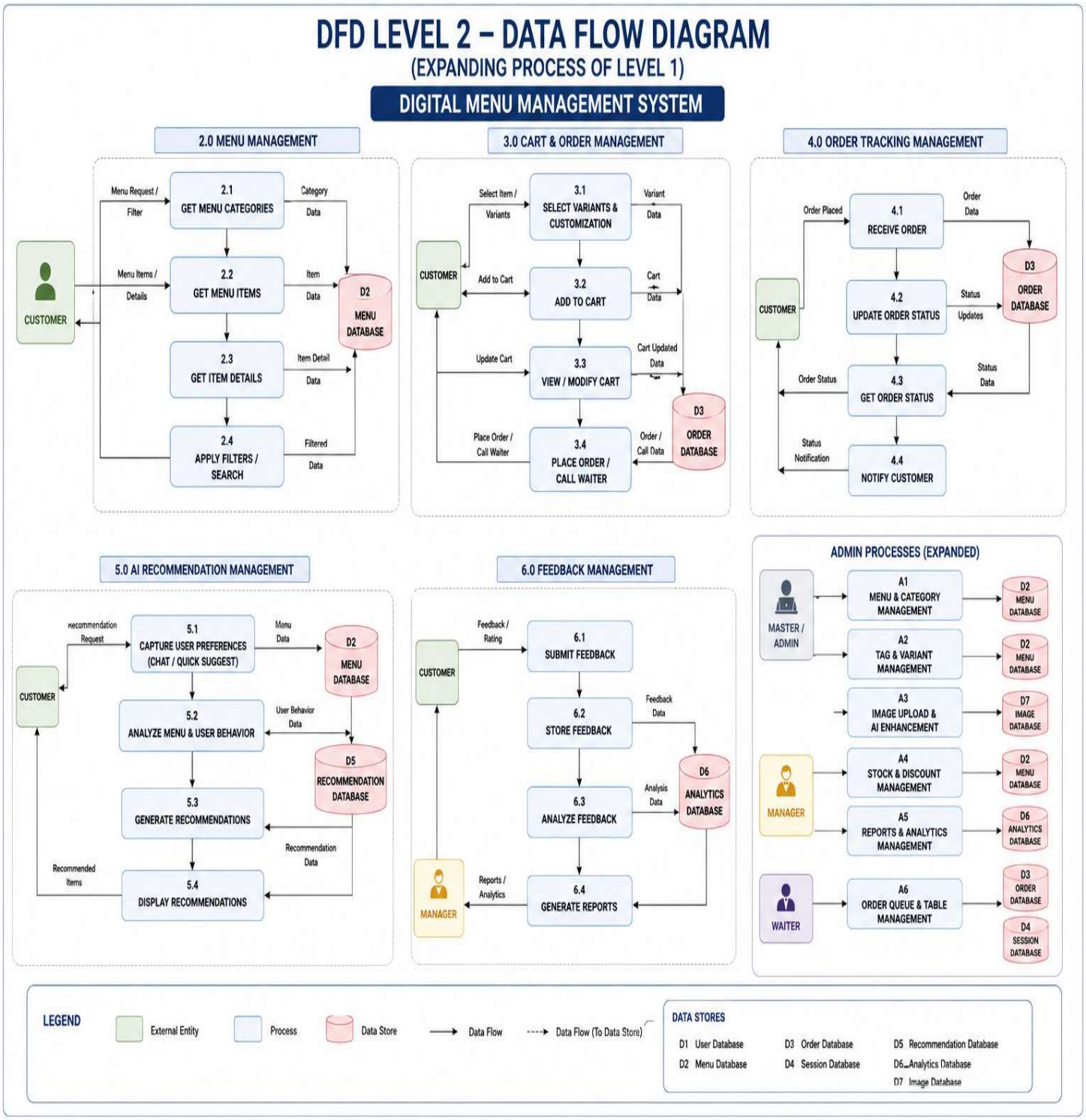


Figure 4: DFD Level 2

5.7 Summary

In this chapter, the Software Requirement Specifications (SRS), system architecture, operating environment, user classes, design constraints, external interface requirements,and data flow structure of the AI-Assisted QR-Based Digital Menu Application were discussed. The chapter explained how different system modules interact to provide QR-based menu accessibility, AI-assisted recommendations, real-time order management, and operational control for modern restaurant environments. The discussed design architecture provides a strong and scalable foundation for implementing an efficient smart restaurant management platform.

Chapter 6

Modeling

This chapter presents the various modeling techniques used to represent the structure, workflow, and interactions within the AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management. Modeling techniques help in understanding how customers, restaurant staff, and administrators interact with the system and how the system processes different operational activities such as QR-based menu access, AI-assisted recommendations, order handling, and restaurant management operations.

The modeling process provides a clear visualization of system functionality, internal workflow, user interaction, and communication between different modules. UML-based diagrams are used to represent system behavior, relationships, and activity flow. These diagrams help developers, administrators, and stakeholders understand the complete operational architecture of the proposed smart restaurant management platform.

6.1 ER Diagram

The Entity Relationship (ER) Diagram is a conceptual database design model used to represent the entities, attributes, and relationships within the AI-Assisted QR-Based Digital Menu Application. It provides a structured view of how data is stored, organized, and interconnected in the system database.

In this project, the ER Diagram illustrates the relationships between various entities such as Users, Tables, Categories, Products, Orders, Order Items, Tags, Product Tags, Offers, Favorites, Feedbacks, and Waiter Calls. The diagram defines primary keys, foreign keys, and cardinality constraints to ensure data integrity and efficient database management.

The ER Diagram helps in understanding the overall database architecture by showing how restaurant operations, menu management, customer interactions, order processing,promotional offers, feedback collection, and waiter assistance functionalities are integrated within a centralized database system. It serves as the foundation for implementing the relational database schema and supports efficient data retrieval, consistency, and scalability of the application.

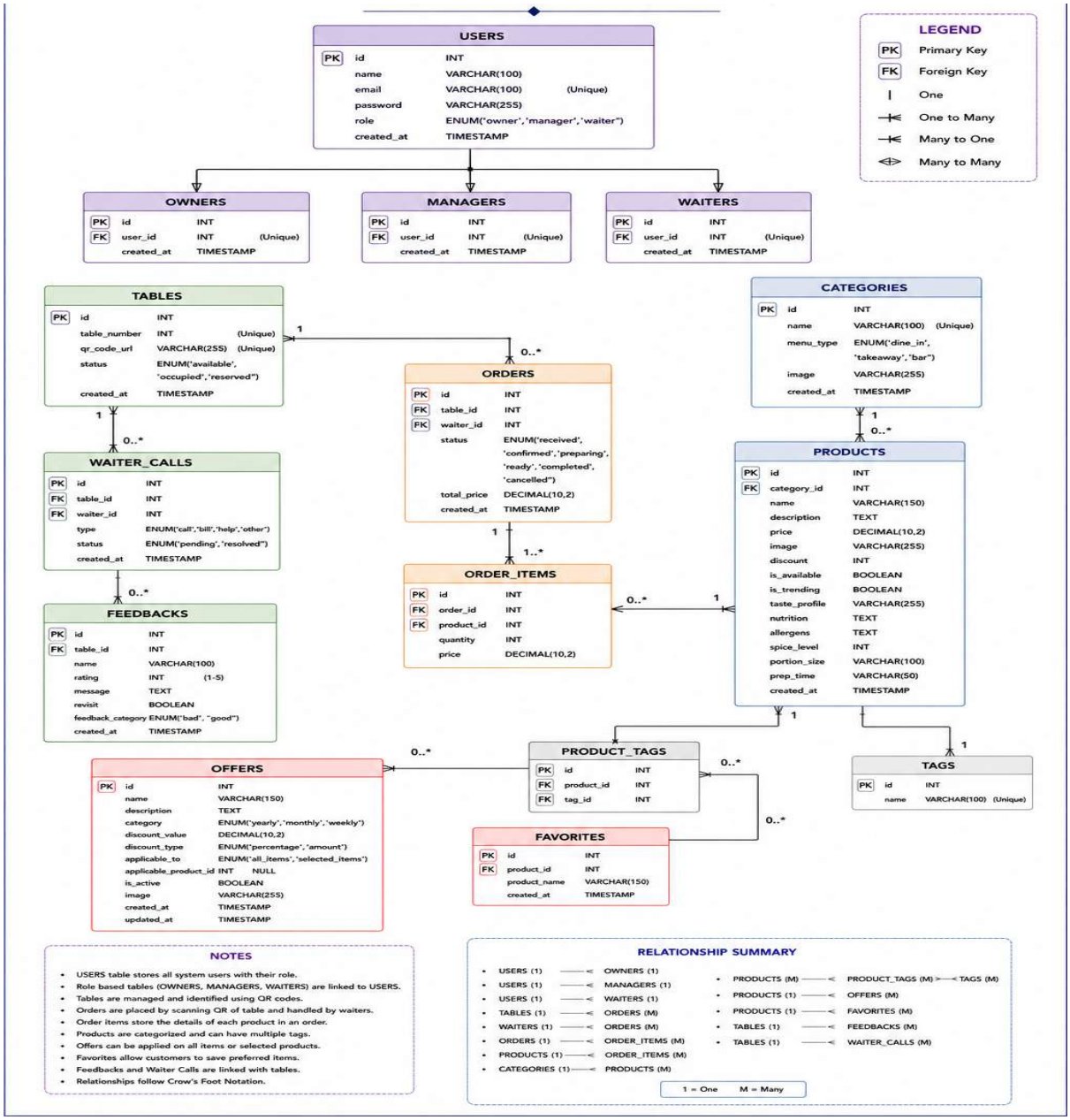


Figure 5: ER Diagram of AI-Assisted QR-Based Digital Menu Application

6.2 Use Case Diagram

A Use Case Diagram is a behavioral UML diagram used to represent the interaction between users (actors) and the system. It provides a graphical overview of the functionalities offered by the AI-Assisted QR-Based Digital Menu Application and illustrates how different users achieve their goals through system features and operational modules.

In this project, the Use Case Diagram represents interactions between Customers, Waiters, Managers, Restaurant Admins, and the digital menu system. It includes functionalities such as QR-based menu access, food browsing, AI-assisted recommendations, cart management, order placement, waiter communication, menu management, analytics monitoring, and operational control.

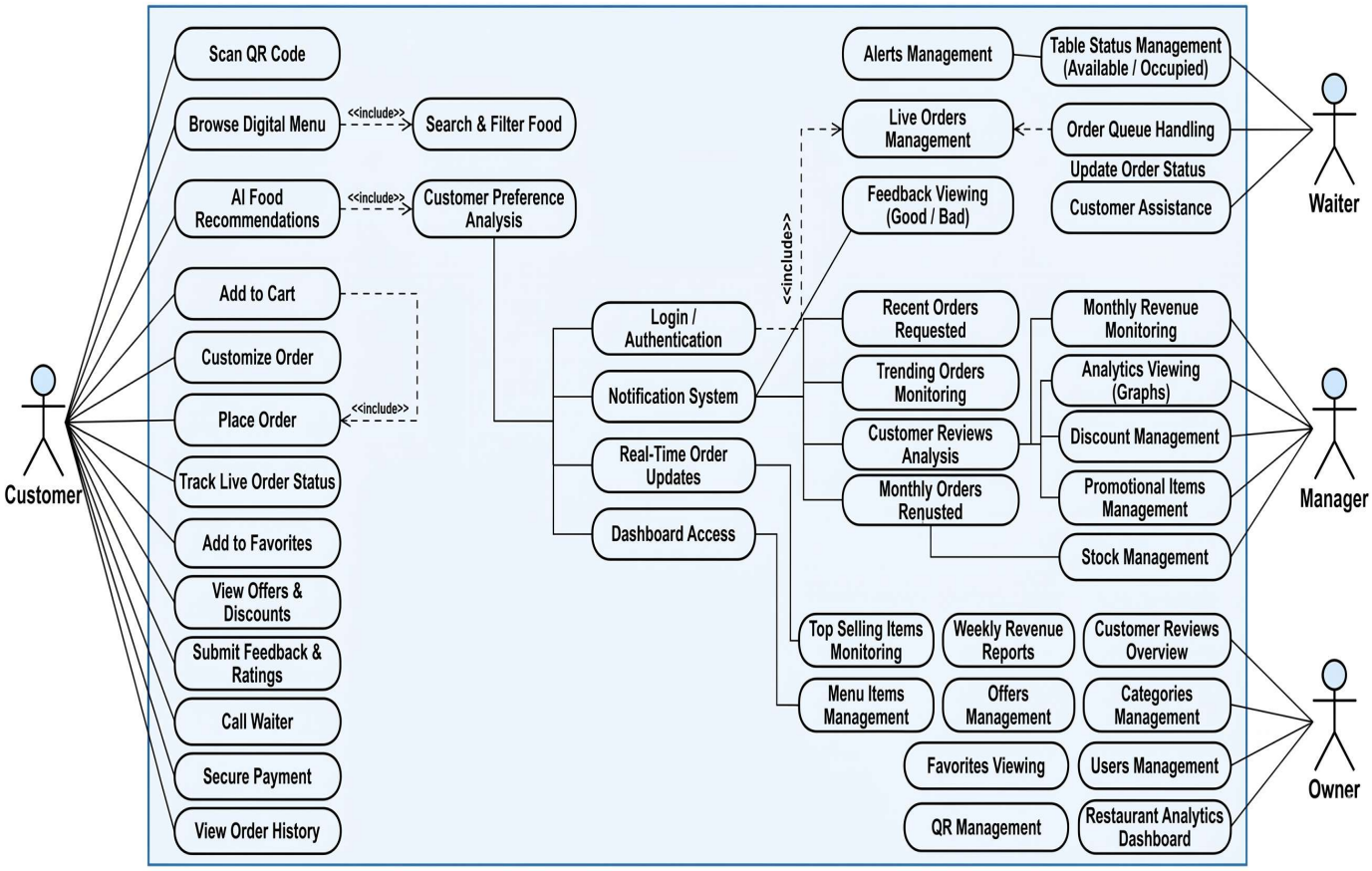


Figure 6: Use Case Diagram of AI-Assisted QR-Based Digital Menu Application

6.3 Class Diagram

The Class Diagram represents the static structure of the AI-Assisted QR-Based Digital Menu Application by showing the major system classes, their attributes, methods, and relationships. It acts as a blueprint for object-oriented system development and demonstrates how different modules interact internally to perform restaurant management operations.

The Class Diagram includes major classes such as Customer, Waiter, Manager, Admin,Menu, Cart, Order, RecommendationEngine, NotificationSystem, PaymentHandler, and DatabaseManager. The diagram represents associations, aggregations, and dependencies between these components to ensure modular and scalable architecture implementation.

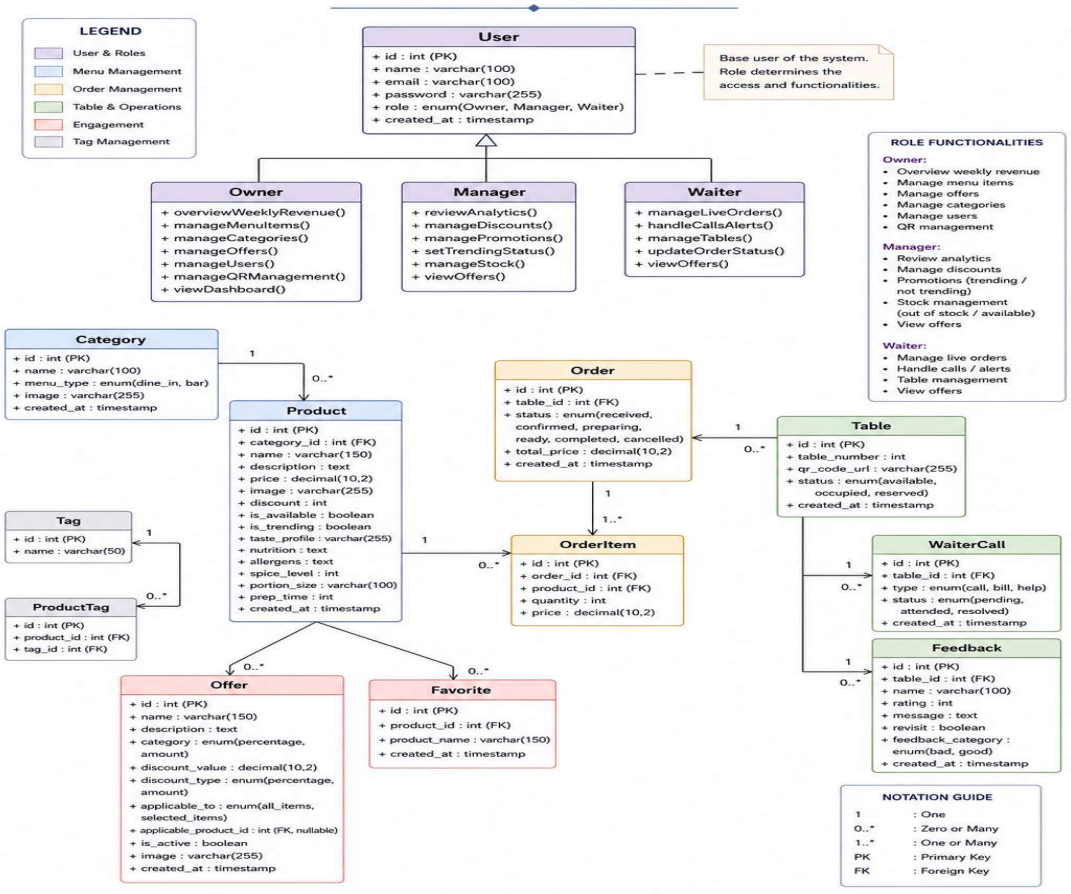


Figure 7: Class Diagram of AI-Assisted QR-Based Digital Menu Application

6.4 Activity Diagram

The Activity Diagram represents the workflow of the AI-Assisted QR-Based Digital Menu Application from QR code scanning to order completion. It illustrates how customers view the menu, receive AI-based recommendations, add items to the cart, and place orders through the system.

The diagram also shows order processing, real-time status updates, waiter notifications, and order confirmation. Decision nodes are used to represent conditions such as item availability and customer actions, providing a clear view of the system's operational flow.

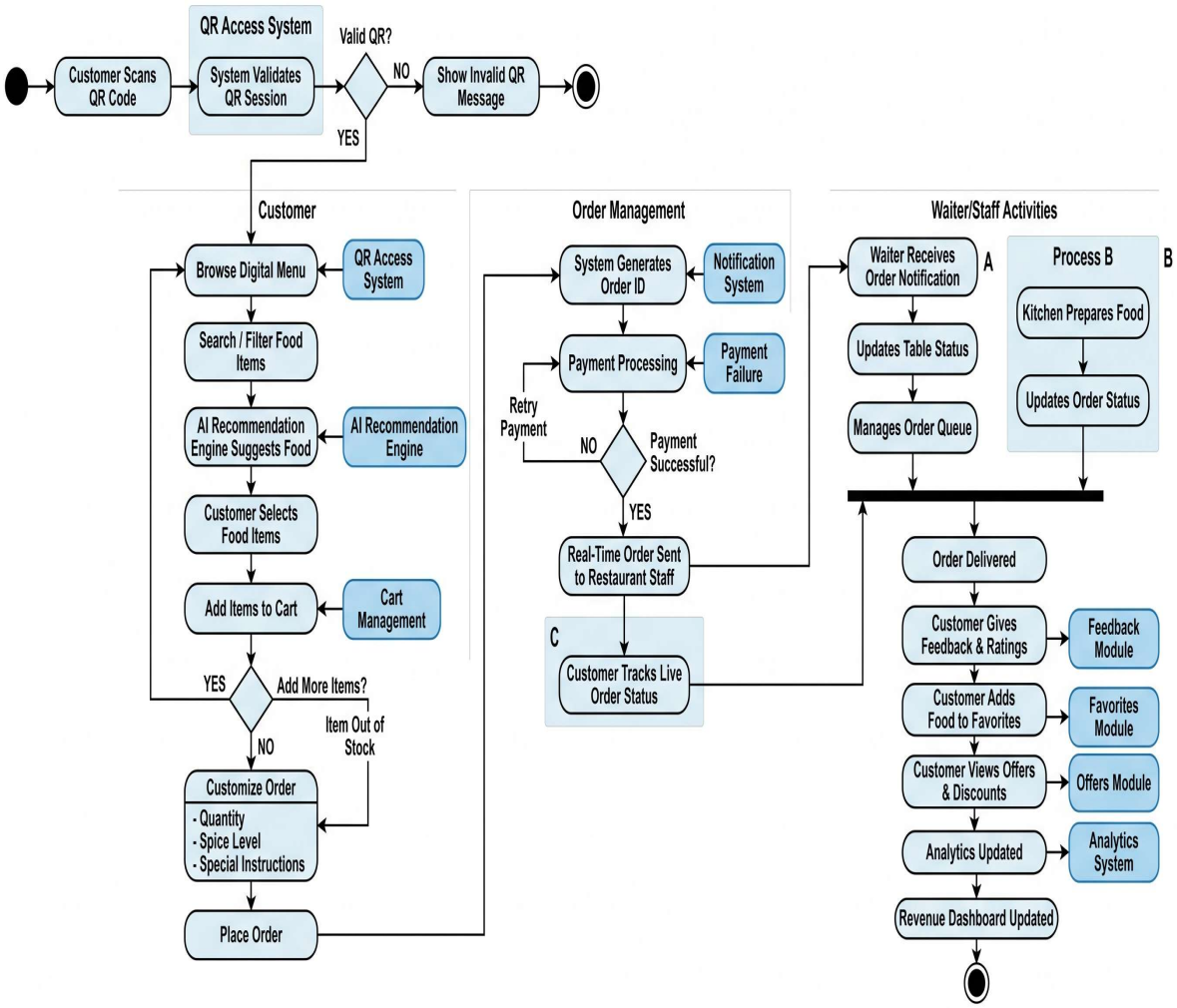


Figure 8: Activity Diagram of AI-Assisted QR-Based Digital Menu Application

6.5 Sequence Diagram

The Sequence Diagram shows the interaction flow between key components of the AI-Assisted QR-Based Digital Menu Application, including the Customer, Frontend Interface, Backend Server, Database, AI Recommendation Engine, and Restaurant Staff. It explains how system modules communicate step by step during a customer's order journey.

The process starts when the customer scans the QR code, browses the menu, receives AI-based recommendations, and places an order. The backend handles requests, updates the database, and sends real-time notifications to restaurant staff for efficient order processing and management.

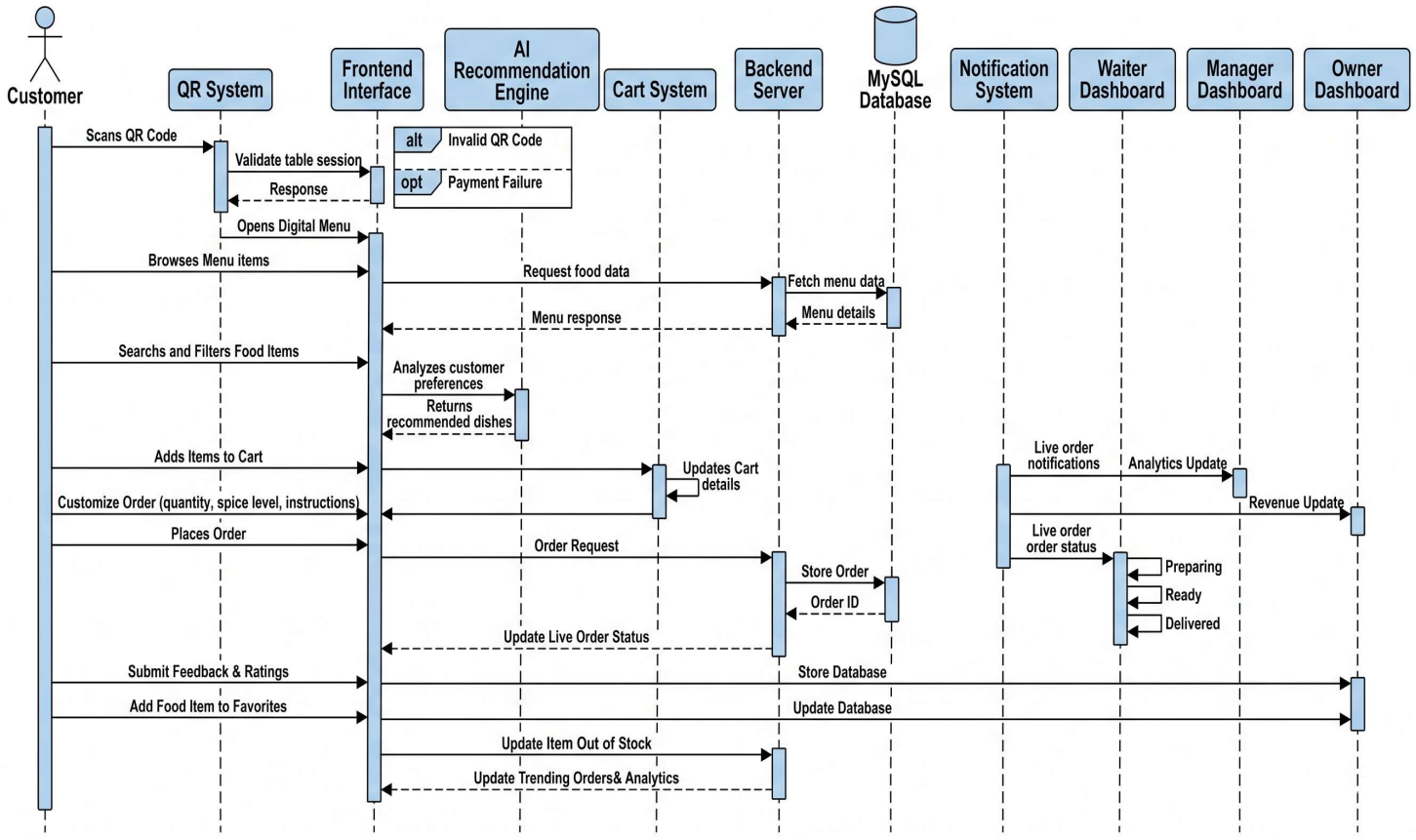


Figure 9: Sequence Diagram of AI-Assisted QR-Based Digital Menu Application

6.6 Summary

In this chapter, various UML-based modeling techniques used for the AI-Assisted QR-Based Digital Menu Application were discussed. The Use Case Diagram represented interactions between customers, restaurant staff, and system functionalities. The Class Diagram illustrated the structural relationship between system modules and operational entities, while the Activity Diagram explained the workflow of customer interactions and restaurant management processes. These modeling techniques provide a clear understanding of the system design and operational architecture required for implementing a scalable and intelligent smart restaurant management platform.

Chapter 7

Implementation & Result

7.1 Implementation Details

The AI-Assisted QR-Based Digital Menu Application is implemented as a modern web-based restaurant management and customer interaction platform that combines QR-based accessibility, AI-assisted food recommendation, real-time order management, and role-based operational control into a single intelligent system. The application is developed using a modular architecture consisting of multiple interconnected components including User Interface Module, QR Access Module, AI Recommendation Module, Cart and Order Management Module, Favorites and Offers Module, Backend Processing Module, Feedback Module, Dashboard Management Module, and Database Management Module.

7.1.1 User Interface Module

This module provides the main interaction layer for customers and restaurant staff through a responsive web application developed using React.js and Tailwind CSS. Customers can instantly access the digital menu by scanning a QR code placed on restaurant tables. The interface supports category-based browsing, smart search, advanced filtering, personalized recommendations, cart management, and real-time order tracking.

The platform also provides separate dashboards for Waiters, Managers, and Restaurant Owners. Food items are displayed using visually rich food cards containing food images, prices, descriptions, ratings, dietary tags, allergen details, and available offers. The application follows a mobile-first responsive design approach to ensure smooth operation across smartphones, tablets, and desktop devices.

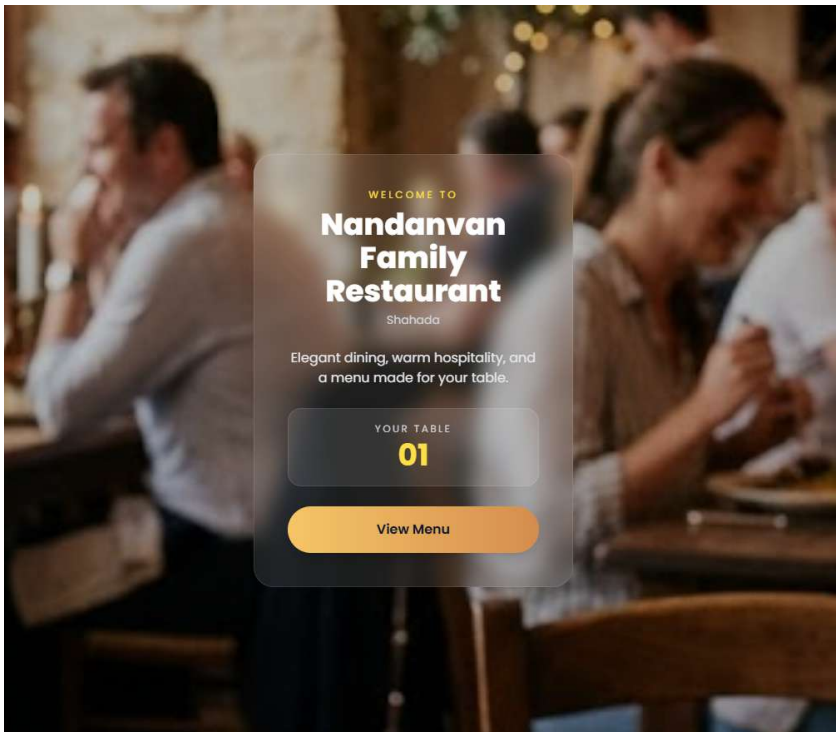


Figure 10: QR Code-Based Restaurant Landing Page

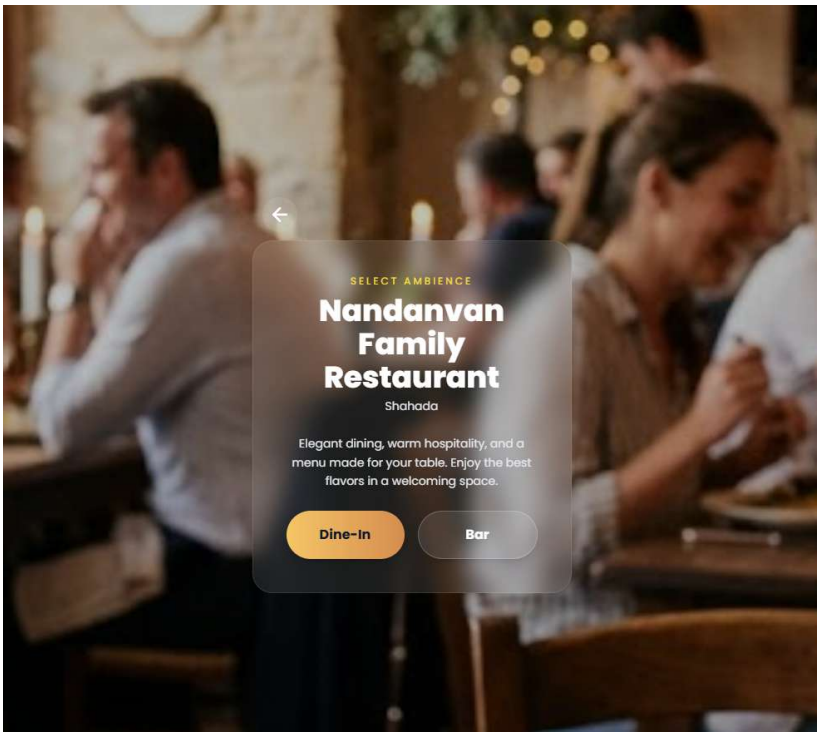


Figure 11: Dining Ambience Selection Interface (Dine-In / Bar Menu)

7.1.2 QR Access Module

The QR Access Module enables customers to access the restaurant menu without downloading any mobile application. Each table contains a unique QR code associated with a specific dining session. When customers scan the QR code using their smartphone camera, the system automatically validates the table session and redirects users to the corresponding restaurant menu interface.

This module improves accessibility, minimizes customer friction, and supports contactless dining experiences. The QR session also helps restaurants manage table-specific ordering and customer sessions efficiently.



Figure 12: QR-Based Menu Access System

7.1.3 AI Recommendation Module

The AI Recommendation Module analyzes customer preferences, browsing behaviour, dietary requirements, favorite dishes, and trending items to generate personalized food suggestions.

Customers receive recommendations based on categories such as Vegetarian, Vegan, Jain, Gluten-Free, Spicy Items, High Protein Meals, Trending Foods, and Chef's Specials. The recommendation engine improves customer engagement and simplifies food discovery by helping users quickly identify dishes according to their preferences.

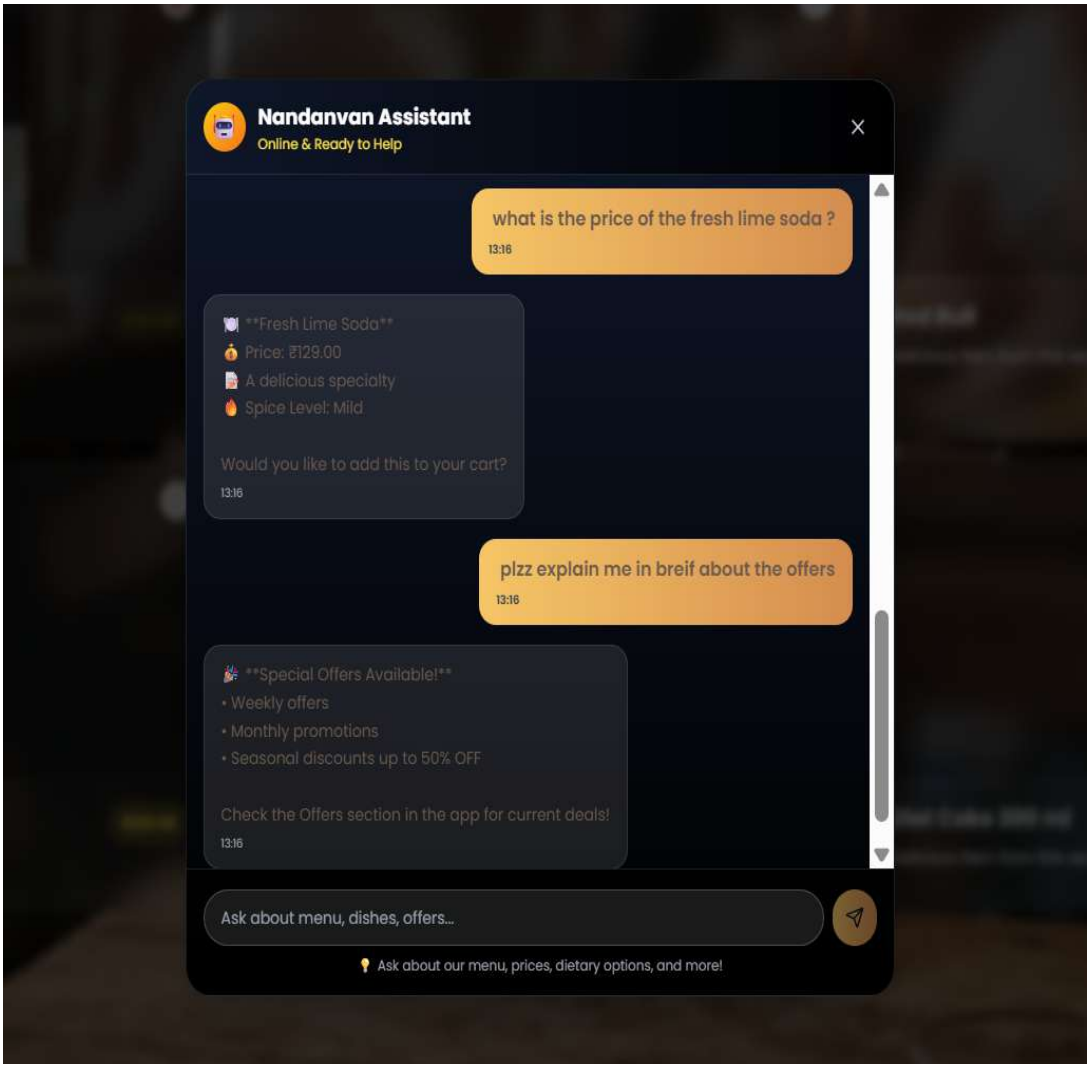


Figure 13: AI Recommendation Assistant Interface

7.1.4 Cart and Order Management Module

The Cart and Order Management Module handles all ordering operations in real time. Customers can add food items to the cart, customize food variants such as size or spice level, adjust quantities, and place orders directly through the application.

The system also provides live order status updates including Order Received, Confirmed, Preparing, Ready, and On The Way. Real-time communication mechanisms are integrated to ensure efficient waiter notifications and smooth coordination between customers and restaurant staff.

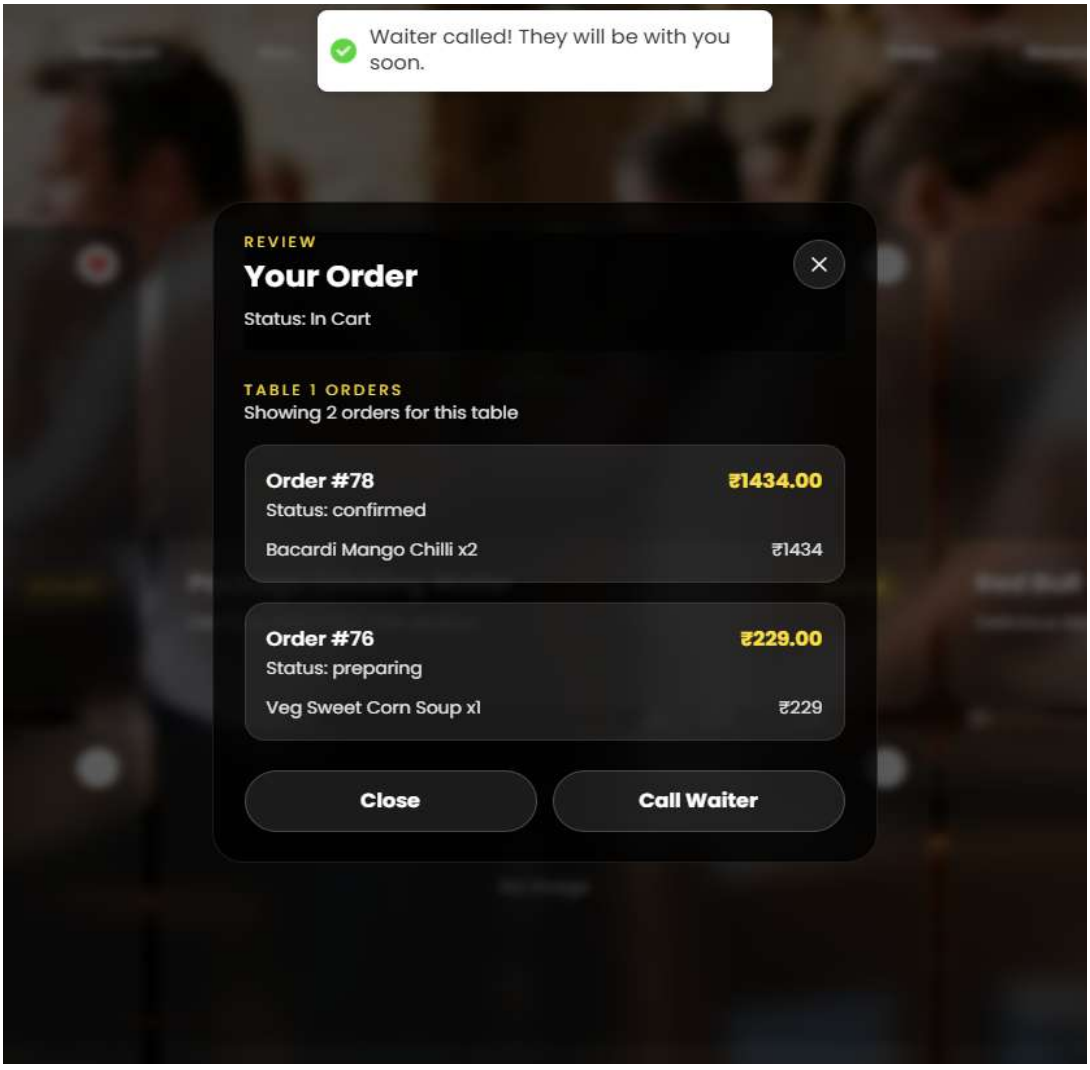


Figure 14: Cart and Real-Time Order Management Interface

7.1.5 Favorites Module

The Favorites Module enables customers to save their preferred dishes for quick and convenient access in future visits. Customers can add menu items to their favorites list and remove them whenever required. This feature enhances the user experience by providing personalized recommendations and reducing the time required to search for frequently ordered items. The module helps customers quickly locate their preferred dishes, making the ordering process more efficient and user-friendly.

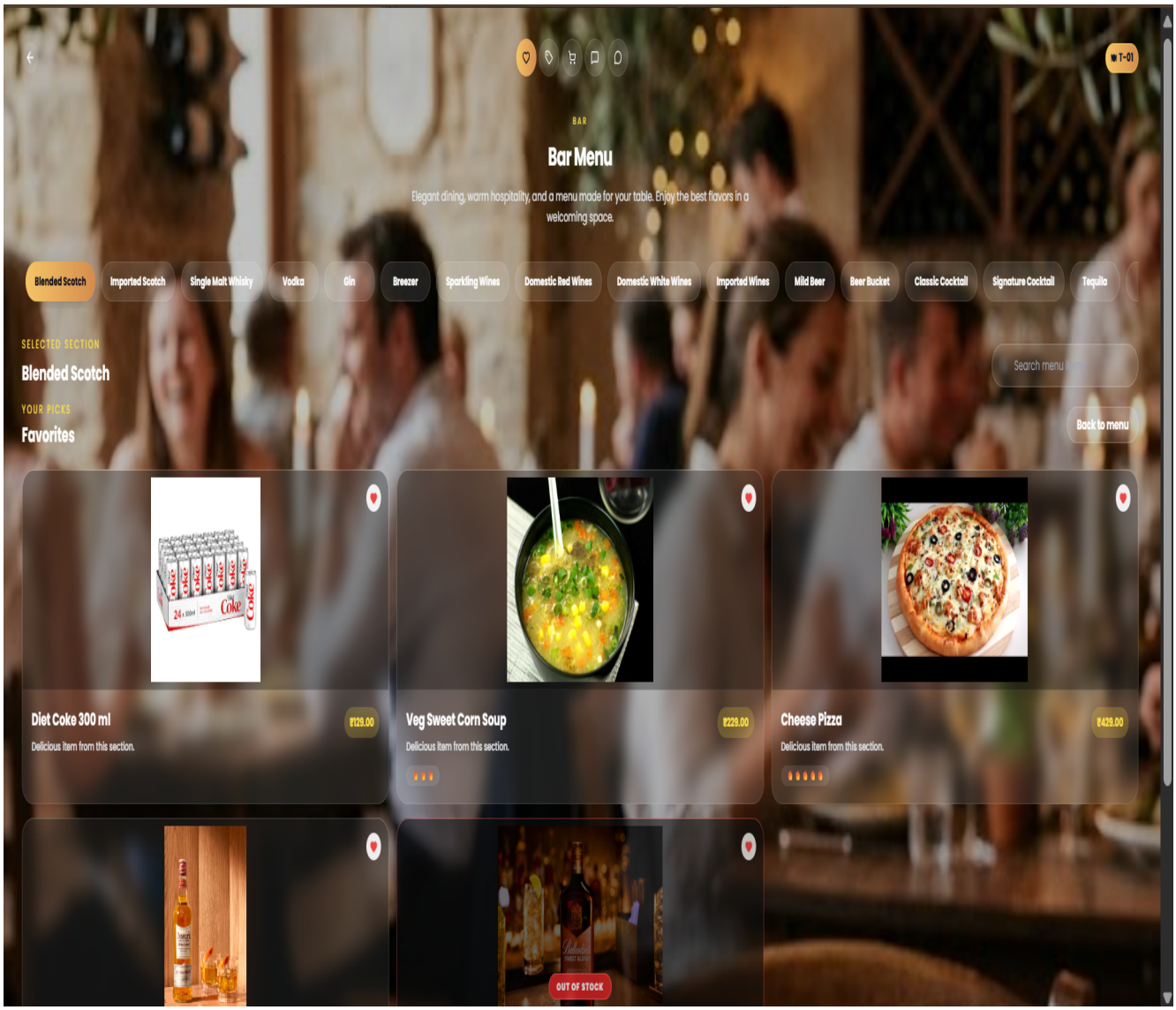


Figure 15: Favorites Module Interface

7.1.6 Offers Module

The Offers Module is designed to manage restaurant promotions, discounts, combo deals, and seasonal campaigns. Customers can view active offers, special discounts, and promotional banners directly while browsing the menu. This feature encourages customer engagement and increases sales by highlighting attractive deals. Restaurant Managers and Owners can create, update, and manage promotional offers through the admin dashboard, ensuring that customers always have access to the latest discounts and marketing campaigns

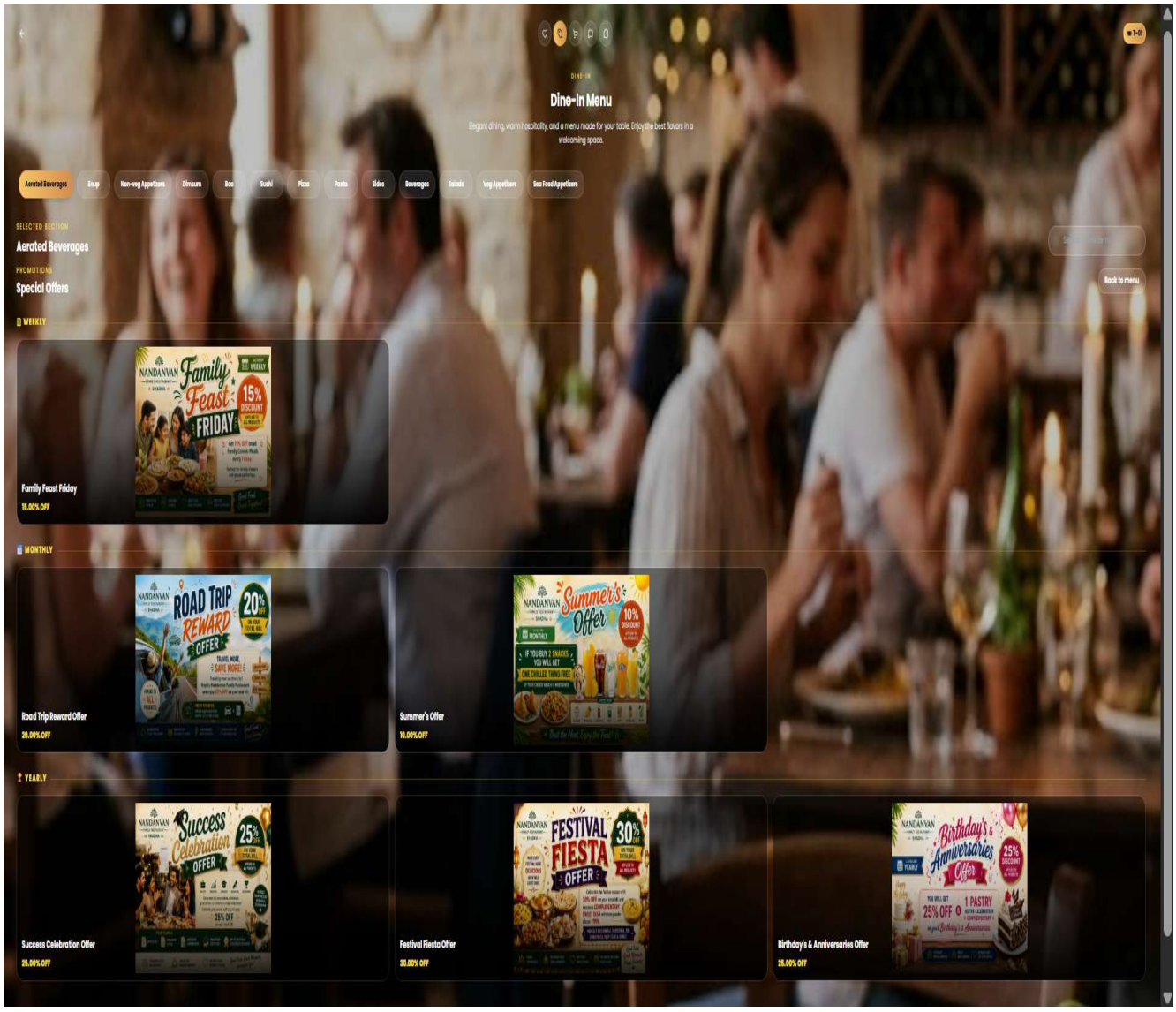


Figure 16: Offers Module Interface

7.1.7 Feedback Module

The Feedback Module allows customers to provide ratings and reviews based on their dining experience. Customers can submit food ratings, restaurant reviews, and service feedback directly through the application interface.

The collected feedback helps restaurant management analyze customer satisfaction levels and identify areas requiring operational improvement. Managers can monitor customer feedback reports and performance metrics using analytics dashboards.

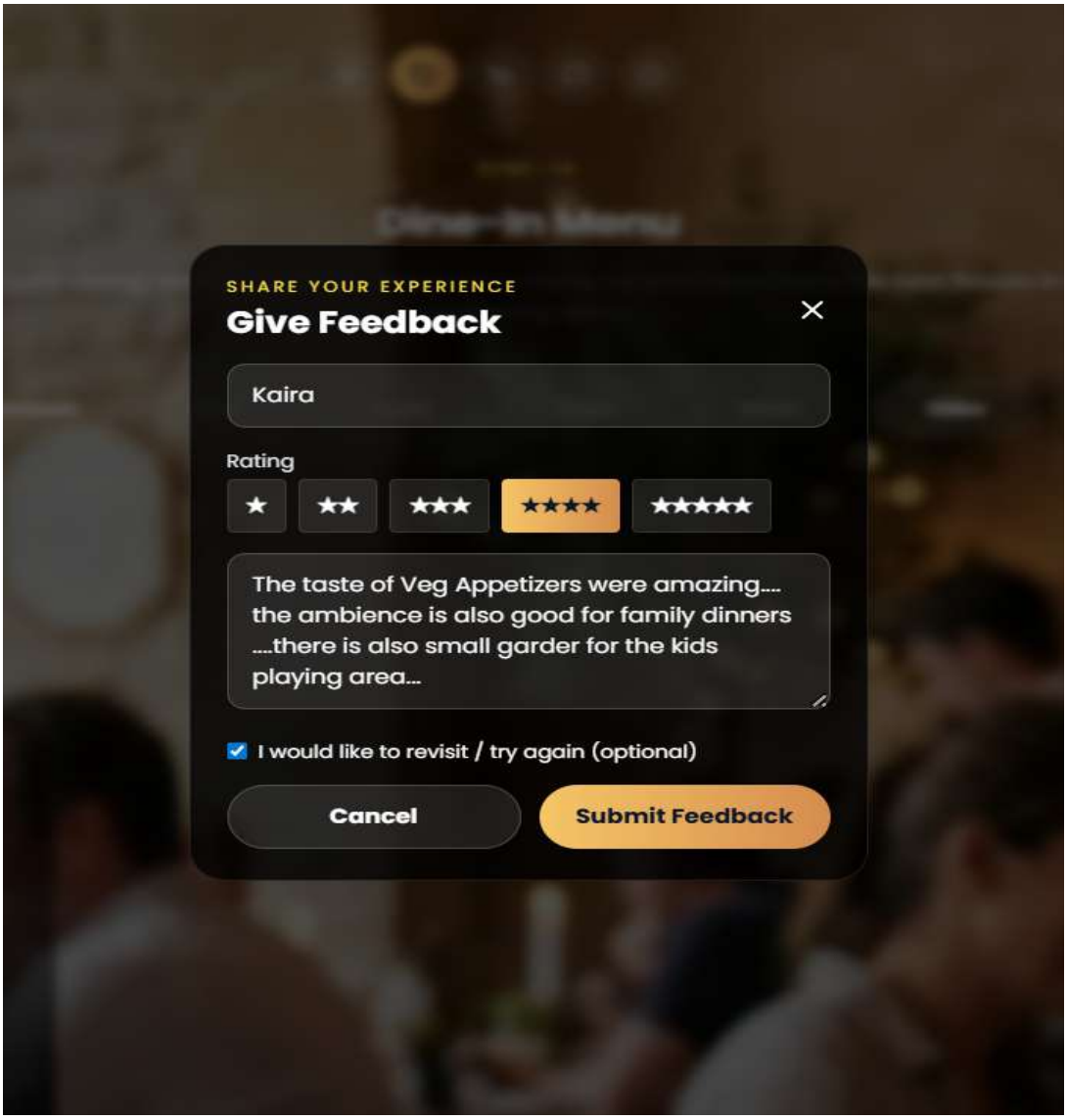


Figure 17: Customer Feedback and Review Interface

7.1.8 Backend Processing Module

The Backend Processing Module is developed using Node.js and Express.js. It manages API communication, authentication, authorization, session handling, real-time order processing, recommendation handling, and database operations.

RESTful APIs are used for frontend-backend communication, while Socket.IO is integrated for live order updates, waiter notifications, and real-time communication handling. The backend also handles menu management, order tracking, analytics generation, and role-based access control.

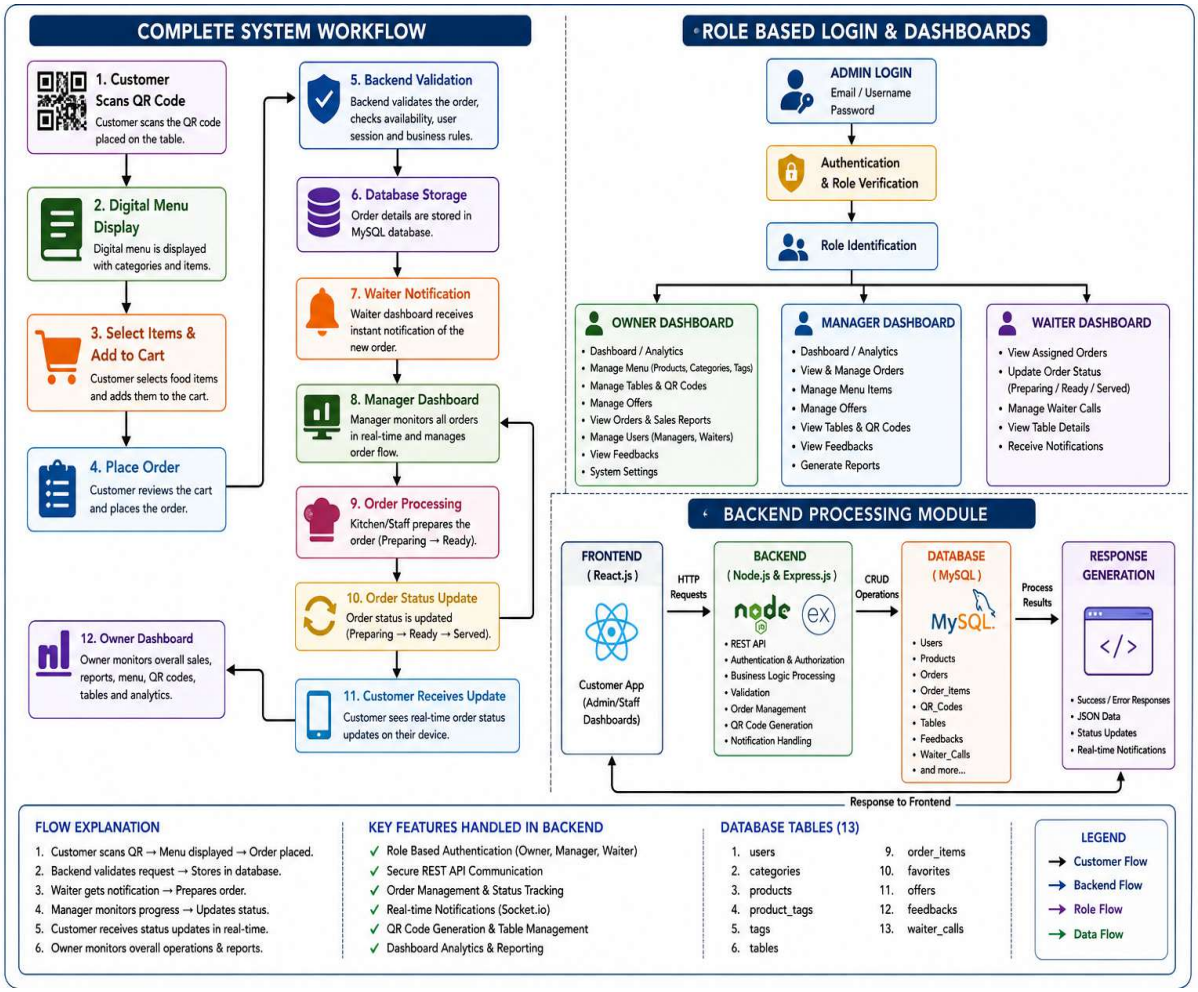


Figure 18: Complete System Workflow and Backend Processing Architecture of the QR-Based Restaurant Management System

7.1.9 Dashboard Management Module

The Dashboard Management Module provides role-based operational dashboards for Waiters, Managers, and Restaurant Owners. Each dashboard is designed according to the responsibilities and access permissions of the respective user role.

7.1.9.1 Waiter Dashboard

The Waiter Dashboard allows restaurant waiters to monitor live customer orders, manage table requests, update order statuses, and respond to "Call Waiter" notifications in real time.

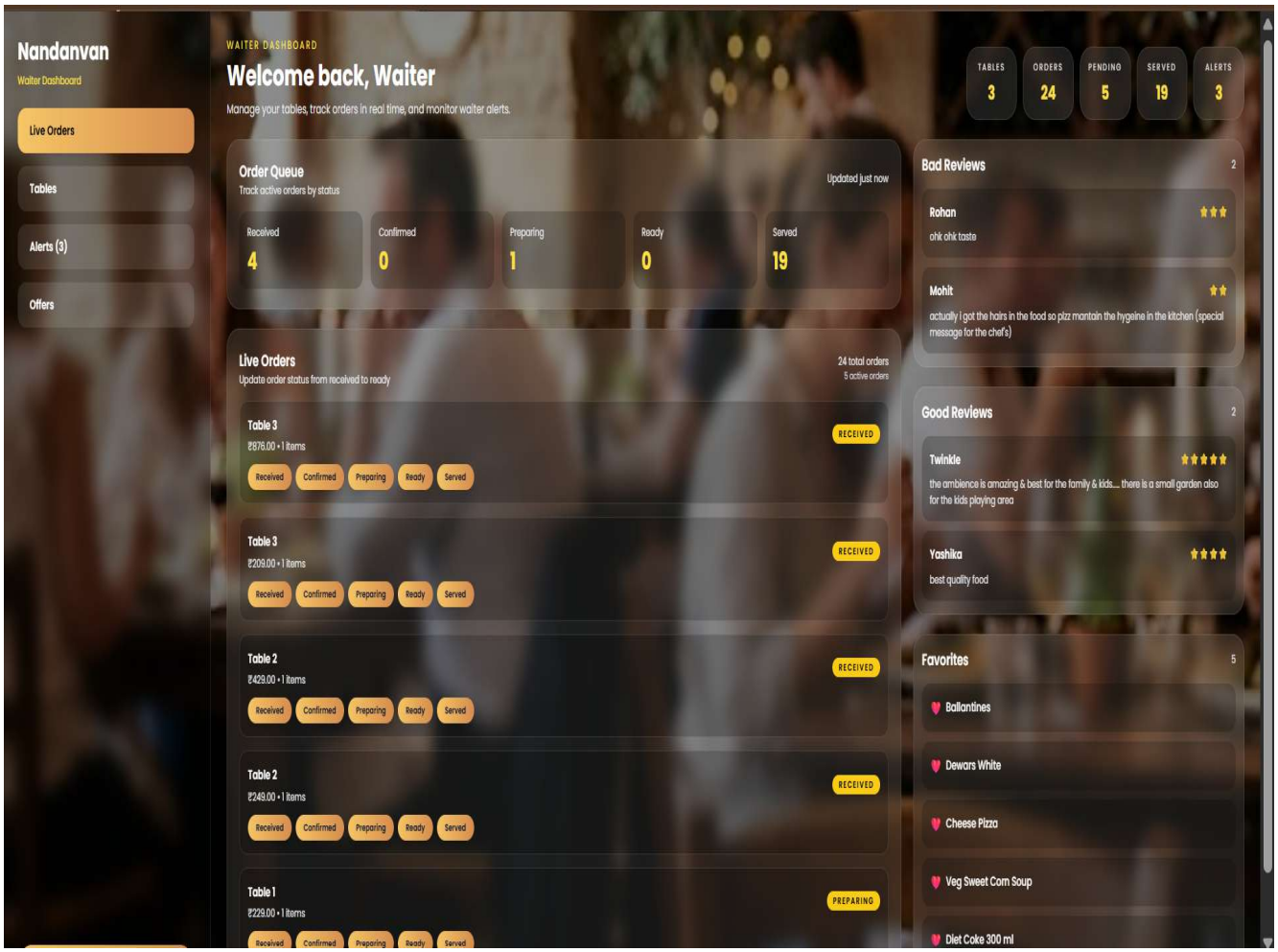


Figure 19: Waiter Dashboard Interface

7.1.9.2 Manager Dashboard

The Manager Dashboard provides tools for monitoring restaurant performance, handling discounts and promotional offers, analyzing customer feedback, monitoring sales activity, and managing stock availability.

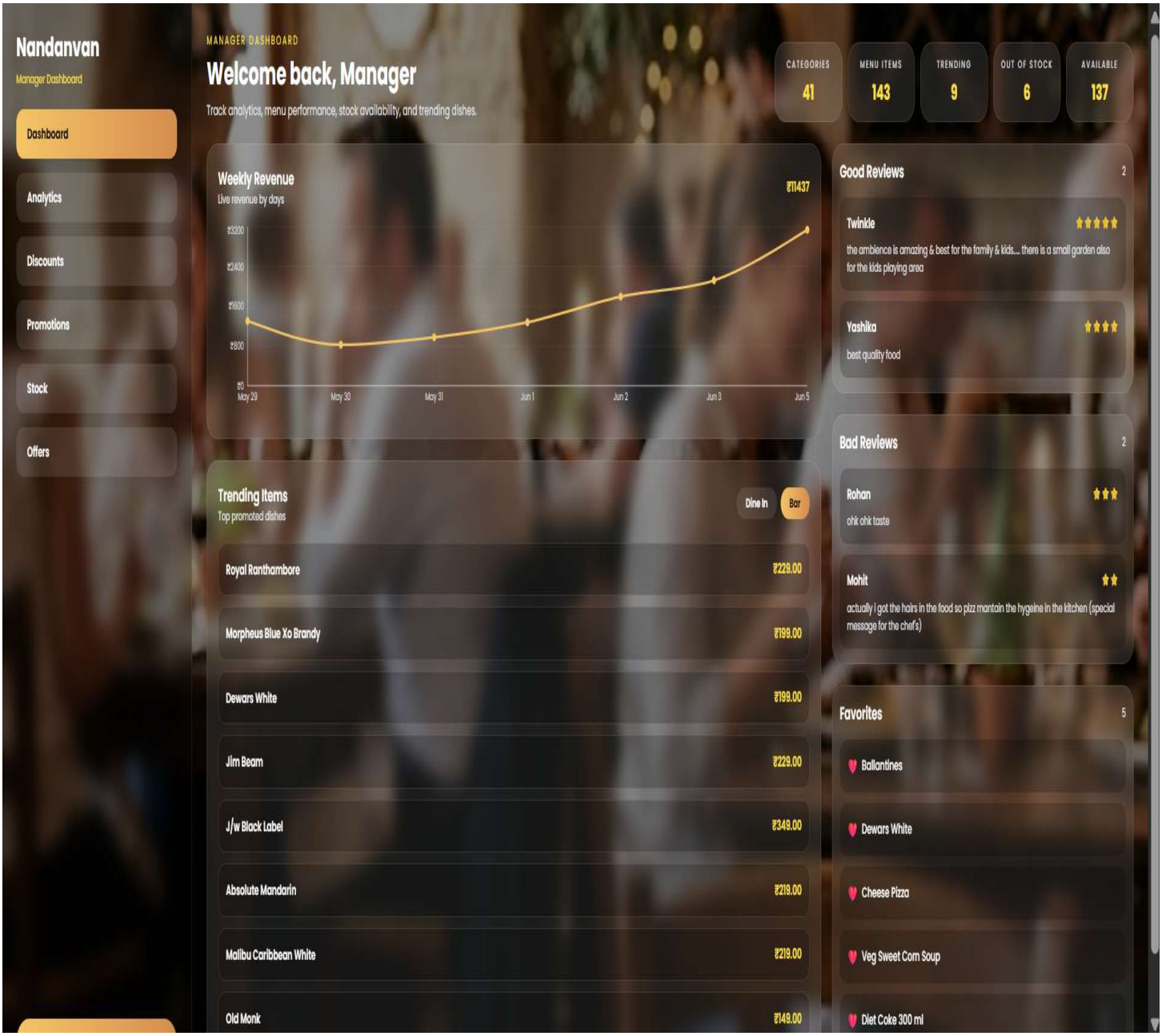


Figure 20: Manager Dashboard Interface

7.1.9.3 Owner Dashboard

The Owner Dashboard provides complete administrative control over the platform. Restaurant Owners can manage menu items, upload food images, manage user roles, configure restaurant settings, monitor business analytics, and supervise overall restaurant operations.

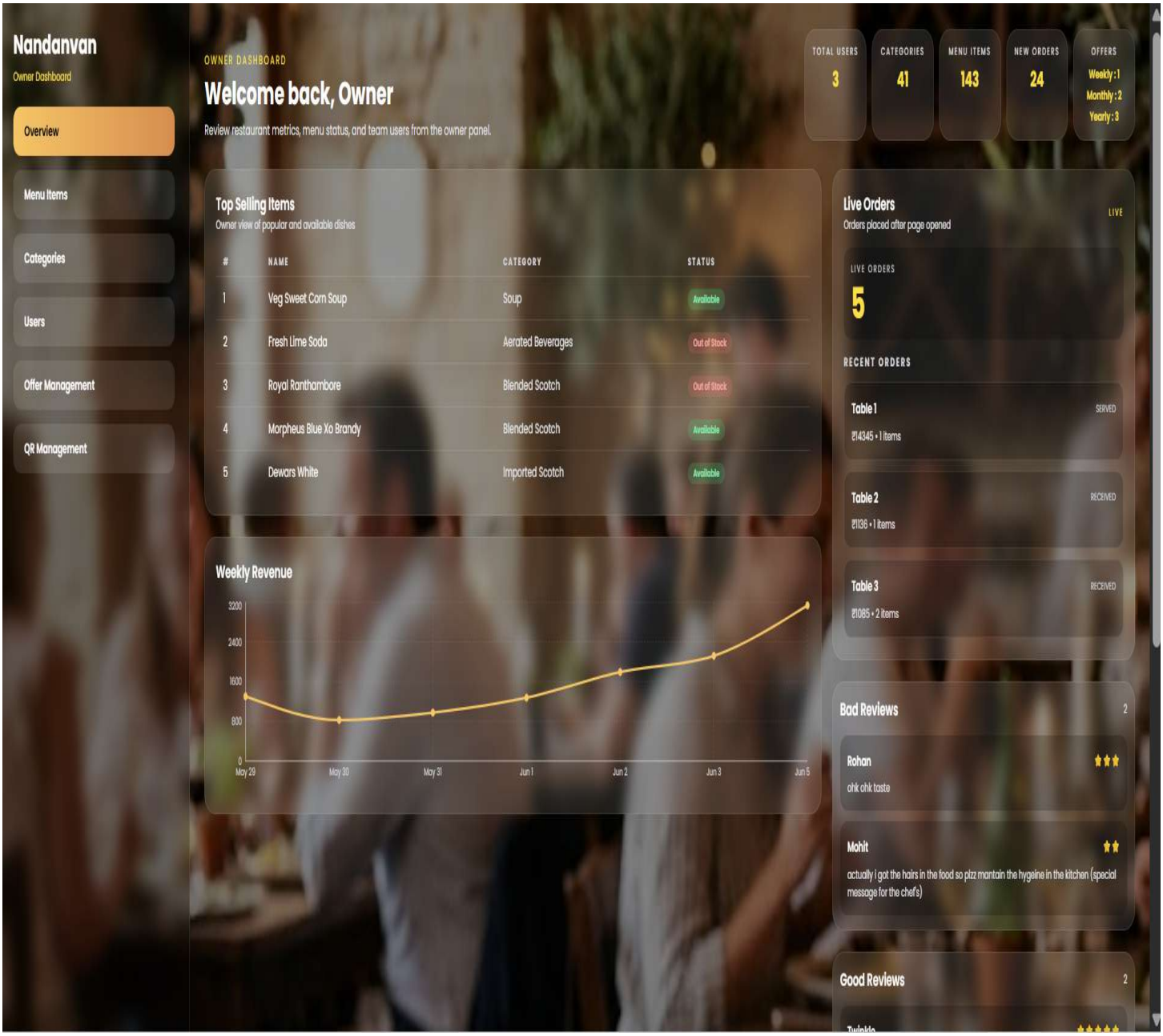


Figure 21: Manager Dashboard Interface

7.1.10 Data Dictionary

The data dictionary provides a detailed description of all database tables used in the QR-Based Digital Restaurant Management System. It defines the structure of each table, including field names, data types, primary keys, foreign keys, and their respective purposes. The database consists of thirteen interconnected tables that support user management, menu management, QR code generation, order processing, feedback collection, offer management, favorites management, and waiter call handling.

1. Users

Field Name	Data Type	Description
id	INT	Unique user ID
name	VARCHAR(100)	User name
email	VARCHAR(100)	User email
password	VARCHAR(255)	Encrypted password
role	ENUM	Owner, Manager, Waiter
created_at	TIMESTAMP	Account creation date

Figure 22: Users Table In Database

2. Categories

Field Name	Data Type	Description
id	INT	Unique category ID
name	VARCHAR(100)	Category name
description	TEXT	Category description
created_at	TIMESTAMP	Creation date

Figure 23: Categories Table In Database

3. Products

Field Name	Data Type	Description
id	INT	Unique product ID
category_id	INT	Category reference
name	VARCHAR(150)	Product name
description	TEXT	Product description
price	DECIMAL(10,2)	Product price
image_url	VARCHAR(255)	Product image
availability	BOOLEAN	Available or not
created_at	TIMESTAMP	Creation date

Figure 24: Products Table In Database

4. Tags

Field Name	Data Type	Description
id	INT	Unique tag ID
name	VARCHAR(100)	Tag name
created_at	TIMESTAMP	Creation date

Figure 25: Tags Table In Database

5. Product_Tags

Field Name	Data Type	Description
id	INT	Unique record ID
product_id	INT	Product reference
tag_id	INT	Tag reference

Figure 26: Product_Tags Table In Database

6. Tables

Field Name	Data Type	Description
id	INT	Unique table ID
table_number	INT	Restaurant table number
status	VARCHAR(50)	Available/Occupied
created_at	TIMESTAMP	Creation date

Figure 27: Tables Table In Database

7. QR_Codes

Field Name	Data Type	Description
id	INT	Unique QR ID
table_id	INT	Table reference
qr_image_url	VARCHAR(255)	QR image location
qr_value	TEXT	Encoded QR data
created_at	TIMESTAMP	Generation date

Figure 28: QR_Codes Table In Database

8. Orders

Field Name	Data Type	Description
id	INT	Unique order ID
table_id	INT	Table reference
user_id	INT	User reference
total_amount	DECIMAL(10,2)	Order total
order_status	VARCHAR(50)	Pending, Preparing, Served, Completed
payment_status	VARCHAR(50)	Paid/Unpaid
created_at	TIMESTAMP	Order date

Figure 29: Orders Table In Database

9. Order_Items

Field Name	Data Type	Description
id	INT	Unique item ID
order_id	INT	Order reference
product_id	INT	Product reference
quantity	INT	Ordered quantity
unit_price	DECIMAL(10,2)	Product price
subtotal	DECIMAL(10,2)	Total item amount

Figure 30: Order_Items Table In Database

10. Favorites

Field Name	Data Type	Description
id	INT	Unique favorite ID
user_id	INT	User reference
product_id	INT	Product reference
created_at	TIMESTAMP	Added date

Figure 31: Favorites Table In Database

11. Offers

Field Name	Data Type	Description
id	INT	Unique offer ID
title	VARCHAR(150)	Offer title
description	TEXT	Offer details
discount_percentage	DECIMAL(5,2)	Discount value
start_date	DATE	Offer start date
end_date	DATE	Offer expiry date

Figure 32: Offers Table In Database

12. Feedbacks

Field Name	Data Type	Description
id	INT	Unique feedback ID
order_id	INT	Order reference
rating	INT	Customer rating
comments	TEXT	Feedback message
created_at	TIMESTAMP	Submission date

Figure 33: Feedbacks Table In Database

13. Waiter_Calls

Field Name	Data Type	Description
id	INT	Unique call ID
table_id	INT	Table reference
call_status	VARCHAR(50)	Pending/Resolved
message	TEXT	Customer request
created_at	TIMESTAMP	Call time

Figure 34: Waiter_Calls Table In Database

7.1.11 Technology Stack

- Frontend: React.js with Tailwind CSS
- Backend: Node.js with Express.js
- Database: MySQL Workbench / MySQL
- Real-Time Communication: Socket.IO
- Authentication: JWT Authentication
- Version Control: Git & GitHub

7.2 Results

The AI-Assisted QR-Based Digital Menu Application successfully demonstrates the integration of QR-based accessibility, AI-assisted recommendations, real-time order management, favorites handling, offers management, feedback collection, dashboard management, and restaurant workflow automation into a unified digital platform.

7.2.1 Functional Results

The application successfully allows customers to scan QR codes, browse interactive menus, search and filter dishes, receive AI-based food recommendations, add items to favorites, view active offers, place customized orders, track live order status, and submit ratings and reviews.

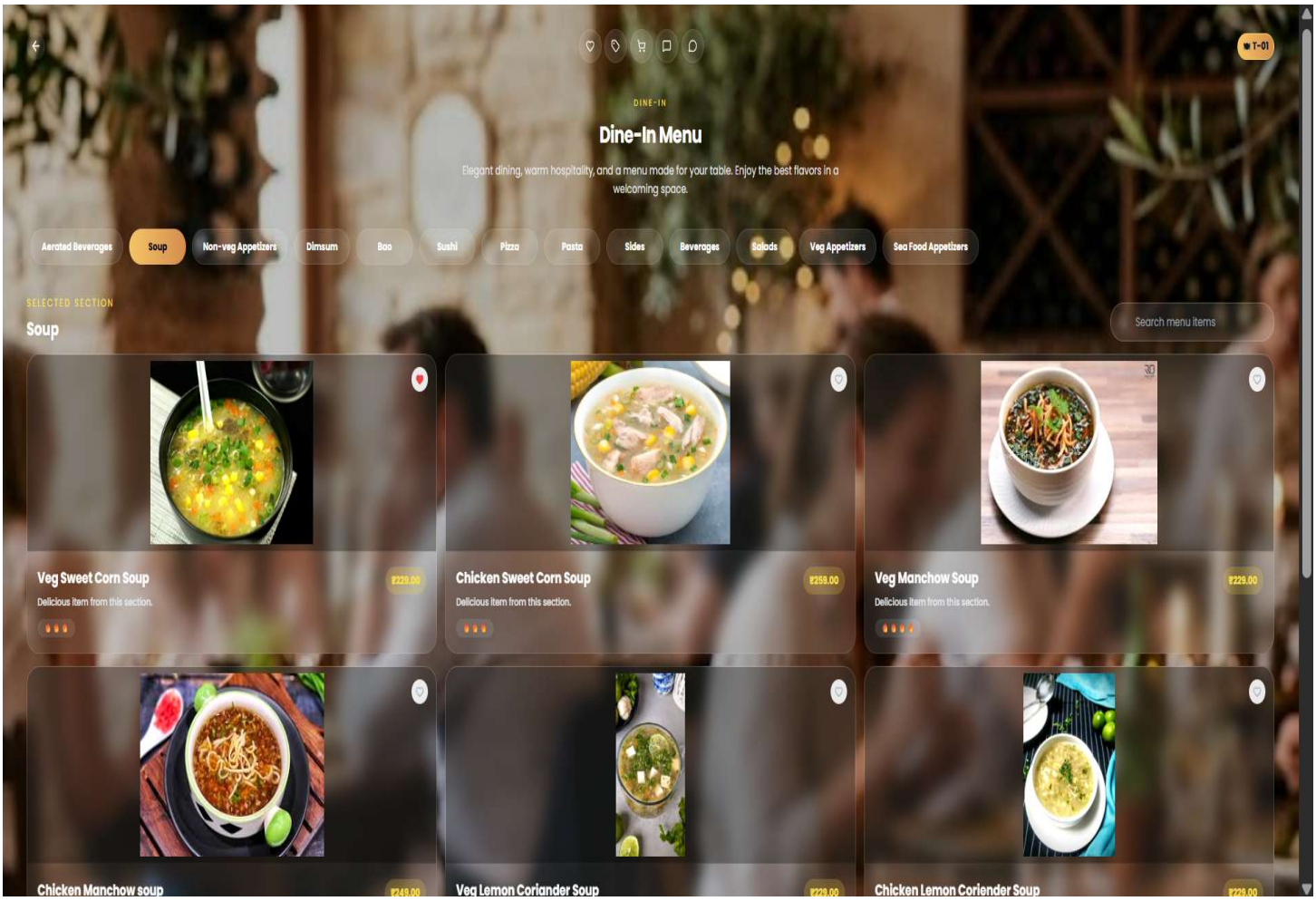


Figure 35: Dine-In-Menu Page

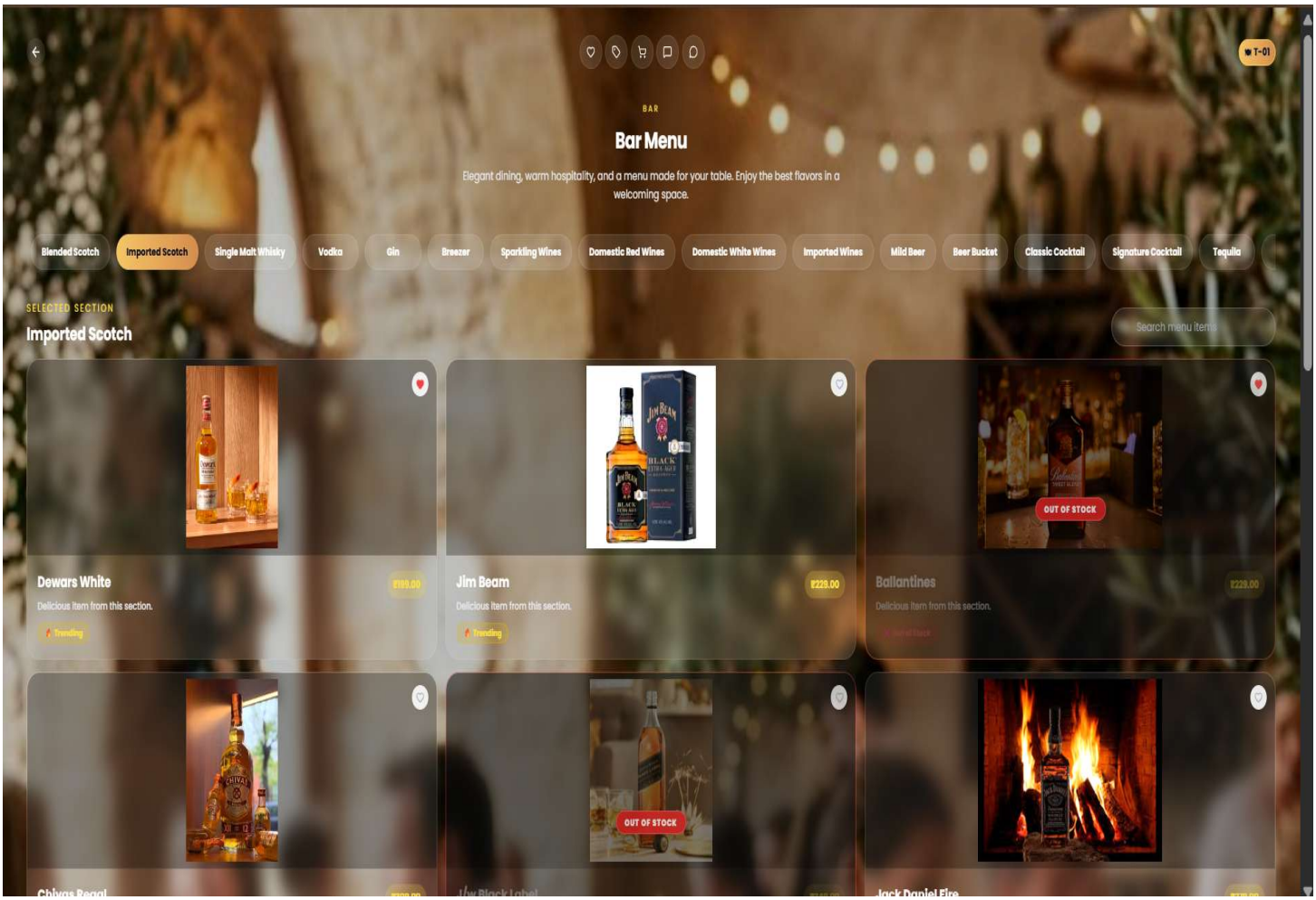


Figure 36: Bar Menu Page

The platform also enables restaurant staff to efficiently manage orders, monitor customer activity, handle table requests, control promotions, and analyze restaurant performance using role-based dashboards.

7.2.2 Performance Results

The system demonstrates smooth real-time communication, responsive UI rendering, and efficient database performance during customer interaction and order processing. The QR-based menu access mechanism significantly reduces customer waiting time and improves restaurant operational efficiency.

The AI recommendation engine successfully enhances customer engagement by suggesting personalized dishes according to user preferences, favorite items, and browsing patterns.

7.2.3 Progress Evaluation

The project modules including QR Access, Digital Menu Management, AI Recommendation, Real-Time Order Processing, Favorites, Offers, Feedback, Dashboard Management, and Role-Based Administration have been successfully implemented and integrated into the final system.

The platform successfully achieves improved customer interaction, reduced order miscommunication, faster restaurant service, efficient operational workflow, and an enhanced digital dining experience for customers and restaurant staff.

7.3 Summary

The AI-Assisted QR-Based Digital Menu Application has been successfully implemented as an intelligent restaurant management and customer interaction platform using modern web technologies and scalable architectural principles. The integration of QR-based accessibility, AI-assisted recommendations, real-time order management, favorites, offers, feedback systems, and role-based dashboards creates a complete digital dining ecosystem for modern restaurants.

The application provides a smart, scalable, visually engaging, and operationally efficient solution that improves customer satisfaction and restaurant management efficiency through intelligent web-based technologies.

Chapter 8

Testing

8.1 Formal Technical Review

Formal Technical Review (FTR) was conducted to ensure that the AI-Assisted QR-Based Digital Menu Application satisfies the specified functional and performance requirements while maintaining high quality in design, implementation, and user experience. The review process involved systematic evaluation of each module including QR-based access, digital menu browsing, AI recommendation system, cart and order management, favorites and offers handling, feedback system, backend processing, and role-based administrative workflow.

During the review process, the complete system architecture, frontend-backend communication, API integration, database operations, and real-time communication mechanisms were analyzed to verify consistency, scalability, responsiveness, and operational correctness. Special attention was given to QR session validation, real-time order synchronization, AI recommendation behaviour, and role-based dashboard functionality to ensure smooth restaurant workflow management across customer and admin interfaces.

Peer reviews and project guide evaluations were conducted to identify interface inconsistencies, order synchronization issues, responsive design problems, and backend communication delays. Feedback received during testing phases helped improve menu loading speed, optimize image rendering, refine order tracking mechanisms, and enhance customer navigation experience throughout the platform.

The review process additionally helped identify potential operational risks such as delayed real-time notifications during high order traffic, improper session handling,

duplicate order submissions, and database synchronization inconsistencies. Corrective measures including optimized API handling, improved session management, lazy image loading, secure authentication mechanisms, and real-time socket communication improvements were implemented to increase overall system reliability and performance.

8.2 Test Plan

The testing of the AI-Assisted QR-Based Digital Menu Application was planned systematically to validate functionality, usability, performance, responsiveness, and reliability under different operational scenarios. Multiple testing approaches were performed to ensure smooth interaction between customer-side and restaurant-side modules.

8.2.1 Objectives of Testing

- To verify successful QR code scanning and table session initialization.
- To ensure proper loading and display of restaurant menu data across all devices.
- To validate AI-assisted food recommendation functionality based on customer preferences and browsing behaviour.
- To verify correct cart management, order placement, and real-time order tracking functionality.
- To ensure proper functionality of favorites, offers, and feedback modules.
- To validate secure role-based access control for Customers, Waiters, Managers, and Restaurant Admins.
- To check system responsiveness, database performance, and real-time communication reliability under different usage conditions.

8.2.2 Testing Strategies

1. Unit Testing

Each module including QR Access, AI Recommendation, Cart Management, Favorites, Offers, Feedback, Authentication, and Database Operations was tested independently to verify correct individual functionality and expected outputs.

2. Integration Testing

Integration testing was performed to ensure smooth communication between frontend interfaces, backend APIs, MySQL database operations, and real-time socket communication mechanisms. Special focus was given to order synchronization, admin dashboard updates, and live notification handling.

3. UI and Responsiveness Testing

The application interface was tested across multiple screen sizes including smartphones, tablets, laptops, and desktop devices to ensure responsive layouts, smooth navigation, and proper visual rendering of menu items and dashboards.

4. Edge Case Testing

The system was tested using challenging operational scenarios such as:

- Invalid or expired QR code access
- Network interruptions during order placement
- Duplicate order submissions
- Empty cart checkout attempts
- Invalid authentication credentials
- High traffic real-time order updates

5. User Scenario Testing

Real-world restaurant interaction scenarios were tested such as:

- Customers scanning QR codes and browsing menu items
- Customers placing customized food orders
- Customers adding dishes to favorites and viewing offers
- Waiters managing live orders and table requests
- Managers monitoring restaurant analytics and order flow
- Customers submitting ratings and dining feedback

8.2.3 Sample Test Cases

Test ID	**Input**	**Expected Output**	**Result**
TC01	Scan valid QR code	Successful table session initialization and menu loading	Pass
TC02	Add food items to cart and place order	Successful order placement and real-time status tracking	Pass
TC03	Apply dietary filters and search items	Accurate filtered menu display	Pass
TC04	Add dishes to favorites	Favorite items stored and displayed successfully	Pass
TC05	View active restaurant offers	Correct promotional offers displayed	Pass
TC06	Submit customer feedback and ratings	Successful review submission and database update	Pass
TC07	Admin updates menu and offers	Updated menu reflected instantly on customer interface	Pass
TC08	Waiter monitors live order requests	Real-time order notifications received successfully	Pass

Table 4 : Sample Test Cases for AI-Assisted QR-Based Digital Menu Application

8.3 Summary

The testing of the AI-Assisted QR-Based Digital Menu Application confirms that the system performs efficiently and reliably across multiple restaurant interaction scenarios. The integration of QR-based access, AI-assisted recommendations, real-time order management, favorites, offers, feedback handling, and role-based dashboards has been successfully validated through structured testing procedures.

The application demonstrates strong performance in customer interaction handling, order synchronization, responsive UI rendering, and secure backend communication. Testing also confirms the system's ability to handle operational edge cases such as invalid QR sessions, network interruptions, and concurrent order processing scenarios.

Overall, the testing phase verifies that the proposed system satisfies its functional, operational, and performance requirements successfully. Future improvements will focus on advanced analytics, multilingual support, enhanced AI personalization, and cloud-based scalability enhancements for large-scale restaurant deployments.

Chapter 9

Technical Specifications

In this chapter, we discuss the applications of the AI-Assisted QR-Based Digital Menu Application and the technical requirements necessary for its implementation, deployment, and operational management. The chapter highlights how the proposed system can improve restaurant management, customer engagement, and operational efficiency through intelligent web technologies and QR-based digital interaction.

9.1 Applications

The AI-Assisted QR-Based Digital Menu Application can be deployed across various restaurant and hospitality environments to enhance customer experience, automate restaurant workflows, and improve service efficiency. The major applications of the proposed system are as follows:

- **Restaurants and Cafes:** Customers can scan QR codes to access interactive digital menus, place orders, view offers, save favorite dishes, and track order status directly from their smartphones without waiting for manual assistance.

- **Hotels and Fine Dining Services:** Hotels and premium restaurants can provide visually immersive dining experiences with personalized food recommendations, real-time order management, and customer feedback systems.

- **Food Courts and Multi-Vendor Dining Areas:** The system can manage multiple restaurant vendors, centralized ordering workflows, and customer sessions efficiently within large dining environments.

- **Cloud Kitchens and Delivery Services:** Restaurant owners can manage menu items, promotional offers, customer reviews, and order tracking through centralized dashboards integrated into the application.

- **Smart Hospitality Management:** The platform enables restaurant managers, waiters, and owners to monitor customer activity, manage stock availability, analyze sales performance, and automate restaurant operations using role-based dashboards.

9.1.1 Hardware Requirements

- Intel/AMD Processor (Minimum Dual Core)
- 2 GB RAM minimum (4 GB recommended)
- Minimum 2 GB Free Storage Space
- Smartphone or Tablet with Camera Access for QR Scanning
- Stable Internet Connection
- Computer or Laptop for Admin Dashboard Management

9.1.2 Software Requirements

- Frontend: React.js with Tailwind CSS
- Backend: Node.js with Express.js
- Database: MySQL Workbench / MySQL
- Real-Time Communication: Socket.IO
- Authentication: JWT Authentication
- APIs: RESTful APIs
- Version Control: Git & GitHub

- Development Tools: VS Code
- Operating System: Windows 10+, Ubuntu, macOS
- Deployment Platform: Vercel / Netlify

9.2 Summary

In this chapter, we discussed the applications and technical specifications of the AI-Assisted QR-Based Digital Menu Application. The system demonstrates practical usability in restaurants, cafes, hotels, cloud kitchens, and smart hospitality environments by improving customer interaction and restaurant workflow automation.

The chapter also presented the hardware and software requirements necessary for successful implementation and deployment of the proposed platform. Overall, the AI-Assisted QR-Based Digital Menu Application provides a scalable, efficient, and modern digital solution for restaurant management using QR-based accessibility, AI-assisted recommendations, real-time communication, and intelligent web technologies.

Chapter 10

Future Scope

The AI-Assisted QR-Based Digital Menu Application successfully integrates QR-based accessibility, AI-assisted recommendations, real-time order management, and restaurant workflow automation into a modern digital dining platform. Although the current system provides a complete solution for restaurant management and customer interaction, several advanced features and technological improvements can be incorporated in future versions to further enhance customer experience, operational efficiency, and scalability.

Future development of the system will focus on improving personalization, expanding smart restaurant functionalities, enhancing AI capabilities, and supporting larger hospitality environments through advanced technologies and intelligent automation.

Future Scope of the Proposed System

- **Multi-Language Support:** Future versions of the application can support multiple regional and international languages such as Hindi, Marathi, and Gujarati to improve accessibility for customers from different linguistic backgrounds.
- **Advanced AI Food Recommendation:** The AI recommendation system can be enhanced using machine learning algorithms that analyze customer ordering history, behavior patterns, seasonal preferences, and real-time trends for more accurate personalized suggestions.
- **Online Payment Integration:** Secure online payment gateways such as UPI, Google Pay, PhonePe, Paytm, and card payment systems can be integrated directly into the platform for seamless digital transactions.

- **Table Reservation System:** Future implementation may include online table booking and reservation management features for reducing waiting time and improving restaurant planning.

- **Loyalty and Reward Programs:** Restaurants can introduce customer loyalty points, membership systems, discount coupons, and reward-based engagement features to increase customer retention.

- **Voice-Based Ordering System:** AI-powered voice ordering functionality can be added to allow customers to place food orders using voice commands for a more interactive dining experience.

- **Smart Analytics and Sales Prediction:** Advanced analytics dashboards and AI-based prediction systems can help restaurant owners analyze customer trends, peak sales periods, and inventory requirements more effectively.

- **Customer Feedback Intelligence:** Future versions may include sentiment analysis on customer reviews and ratings to automatically identify customer satisfaction levels and operational improvement areas.

The future enhancements of the AI-Assisted QR-Based Digital Menu Application will further improve restaurant automation, customer satisfaction, and intelligent decision-making capabilities. The proposed system has strong potential to evolve into a complete smart restaurant ecosystem using advanced AI technologies, cloud infrastructure, and real-time digital services.

Chapter 11

Conclusion

The AI-Assisted QR-Based Digital Menu Application successfully demonstrates how QR-based accessibility, AI-assisted recommendations, and real-time restaurant management can be integrated into a smart digital dining platform. The system improves customer experience and simplifies restaurant operations through intelligent and interactive web technologies.

The platform allows customers to access digital menus instantly using QR codes and provides features such as personalized food recommendations, smart filtering, favorites management, offers handling, feedback submission, and live order tracking. These functionalities improve customer engagement and make the ordering process faster and more convenient.

The application also provides separate dashboards for Waiters, Managers, and Restaurant Owners/Admins for efficient workflow management and operational control. Real-time order synchronization, customer analytics, and promotional management help restaurants improve service quality and reduce operational delays.

The use of technologies such as React.js, Node.js, Express.js, Tailwind CSS, Socket.IO, JWT Authentication, and MySQL Workbench ensures responsive performance, secure data management, and smooth real-time communication across multiple devices.

Overall, the AI-Assisted QR-Based Digital Menu Application provides a scalable, efficient, and modern solution for smart restaurant management by combining AI-assisted interaction, QR-based accessibility, and advanced web technologies into a unified digital restaurant ecosystem.

Appendix A

Glossary

This section provides definitions of important terms, technologies, and concepts used in the AI-Assisted QR-Based Digital Menu Application for Smart Restaurant Management.

QR Code

A machine-readable code used for instantly accessing the digital restaurant menu through smartphone scanning without requiring application installation.

PWA (Progressive Web Application)

A web-based application that provides app-like functionality directly through a browser while supporting responsiveness, fast loading, and cross-device compatibility.

AI Recommendation System

An intelligent recommendation mechanism that suggests personalized food items based on customer preferences, dietary choices, favorites, and browsing behaviour.

Responsive Design

A UI design approach that automatically adjusts layouts and components according to different screen sizes such as mobile phones, tablets, and desktops.

JWT Authentication

JSON Web Token (JWT) based authentication system used for secure user login, authorization, and session management.

Rest API

Representational State Transfer (REST) API used for communication between frontend and backend modules through HTTP requests and responses.

Socket.IO

A real-time communication library used for live order updates, waiter notifications, and instant dashboard synchronization.

MySQL

A relational database management system used for storing customer details, menu items, orders, offers, favorites, feedback, and restaurant management data.

MySQL Workbench

A graphical database management tool used for designing, managing, and monitoring the MySQL database structure of the application.

Frontend

The client-side portion of the application that users interact with directly through browsers. In this project, React.js and Tailwind CSS are used for frontend development.

Backend

The server-side logic responsible for handling APIs, authentication, order processing, database communication, and business logic operations.

React.js

A JavaScript library used for building interactive and component-based user interfaces for the digital menu platform.

Tailwind CSS

A utility-first CSS framework used for designing responsive and visually attractive UI components efficiently.

Node.js

A JavaScript runtime environment used for backend development and server-side processing.

Express.js

A lightweight Node.js framework used to create RESTful APIs and backend services.

Role-Based Access Control

A security mechanism where different users such as Customers, Waiters, Managers, and Owners receive separate access permissions according to their responsibilities.

Favorites Module

A functionality that allows customers to save preferred dishes for faster access during future visits.

Offers Module

A restaurant promotion management feature used for displaying discounts, combo deals, and seasonal offers to customers.

Feedback Module

A module that allows customers to submit ratings, reviews, and dining feedback for restaurant performance analysis.

Real-Time Order Tracking

A live order monitoring feature that displays statuses such as Order Received, Preparing, Ready, and Delivered.

Admin Dashboard

A centralized management interface used by restaurant owners and managers for handling menus, users, orders, offers, analytics, and restaurant operations.

Session Management

A process used to maintain customer dining sessions and authentication states securely throughout application usage.

Cloud Deployment

Hosting and running the application on online servers to ensure accessibility, scalability, and high availability.

Appendix B

Achievements

- **Real Client-Based Project Development**

  The AI-Assisted QR-Based Digital Menu Application was developed based on real restaurant management requirements received directly from the client, focusing on solving practical dining and operational challenges.

- **Successful Implementation of AI-Assisted Restaurant Platform**

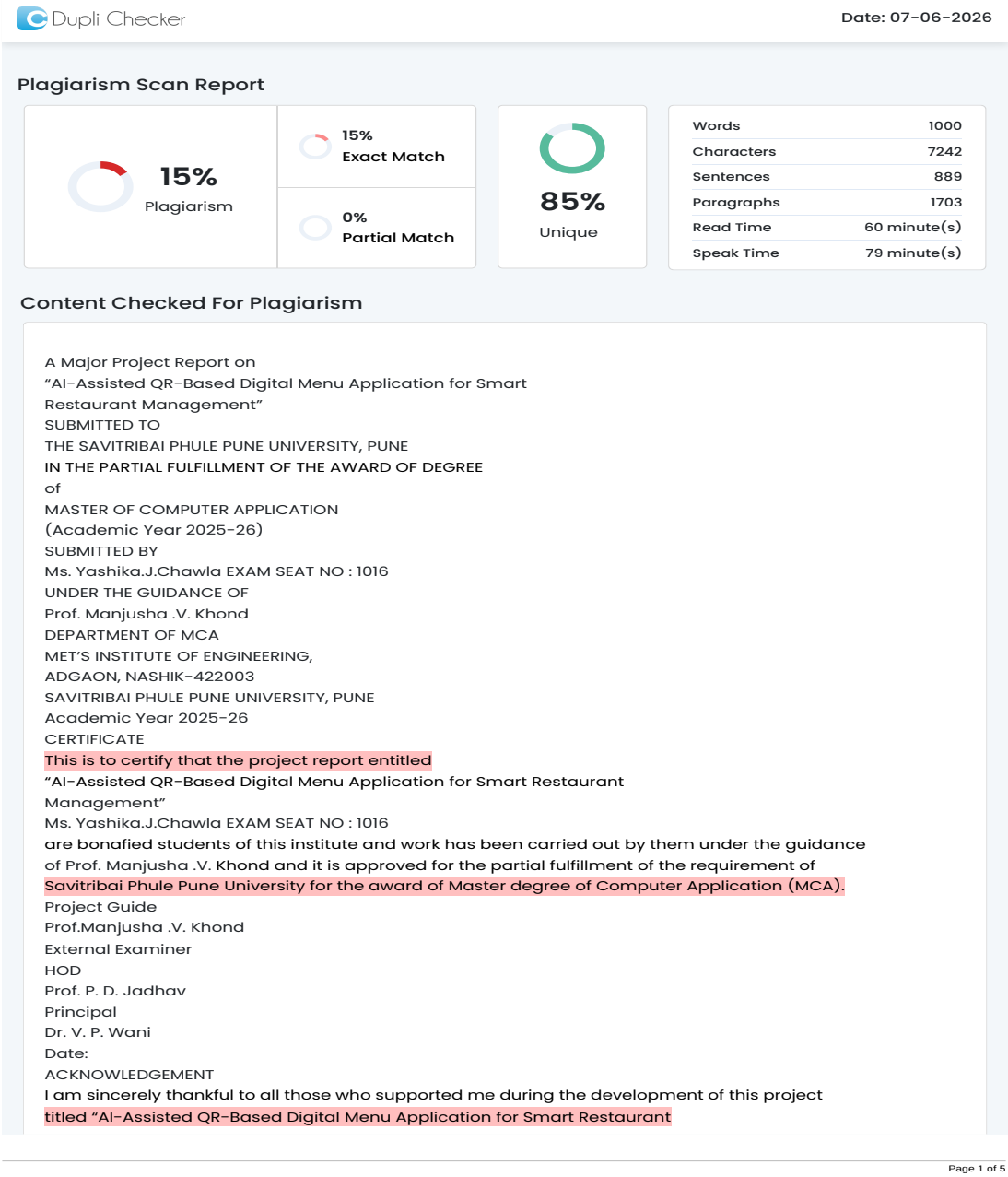
  The project successfully integrates QR-based menu access, AI-assisted food recommendations, real-time order management, favorites, offers, and feedback modules into a single smart restaurant platform.

- **Role-Based Dashboard and Database Management**

  Separate dashboards for Waiters, Managers, and Restaurant Owners were successfully implemented along with secure database management using MySQL Workbench.

Appendix C

Plagiarism Report



Dupli Checker

Date: 07-06-2026

Plagiarism Scan Report

15% Plagiarism

15% Exact Match

0% Partial Match

85% Unique

Words	1000
Characters	7242
Sentences	889
Paragraphs	1703
Read Time	60 minute(s)
Speak Time	79 minute(s)

Content Checked For Plagiarism

A Major Project Report on
"AI-Assisted QR-Based Digital Menu Application for Smart
Restaurant Management"
SUBMITTED TO
THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE AWARD OF DEGREE
of
MASTER OF COMPUTER APPLICATION
(Academic Year 2025-26)
SUBMITTED BY
Ms. Yashika.J.Chawla EXAM SEAT NO : 1016
UNDER THE GUIDANCE OF
Prof. Manjusha .V. Khond
DEPARTMENT OF MCA
MET'S INSTITUTE OF ENGINEERING,
ADGAON, NASHIK-422003
SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
Academic Year 2025-26
CERTIFICATE
This is to certify that the project report entitled
"AI-Assisted QR-Based Digital Menu Application for Smart Restaurant
Management"
Ms. Yashika.J.Chawla EXAM SEAT NO : 1016
are bonafied students of this institute and work has been carried out by them under the guidance
of Prof. Manjusha .V. Khond and it is approved for the partial fulfillment of the requirement of
Savitribai Phule Pune University for the award of Master degree of Computer Application (MCA).
Project Guide
Prof.Manjusha .V. Khond
External Examiner
HOD
Prof. P. D. Jadhav
Principal
Dr. V. P. Wani
Date:
ACKNOWLEDGEMENT
I am sincerely thankful to all those who supported me during the development of this project
titled "AI-Assisted QR-Based Digital Menu Application for Smart Restaurant

Page 1 of 5

Figure 37: Plagiarism Scan Report (1 of 5) – Overall Similarity Summary

Management". Their guidance and encouragement helped me complete this work successfully.

I would like to sincerely thank my internal guide, Prof. Manjusha V. Khond, for her constant support and valuable suggestions throughout the project development. Her guidance helped me understand the requirements clearly and improve the implementation of the system.

I am grateful to the Head of the Department (MCA), Prof. P. D. Jadhav, and the Project Coordinator, Prof. Sonali L. Vidhate, for their support, guidance, and for providing the required resources to complete this project smoothly.

I would also like to thank all faculty members of the MCA Department for their help and guidance during this project. I am also thankful to my classmates, friends, and family members for their continuous encouragement and support.

Finally, I acknowledge everyone who supported me directly or indirectly in completing this project.

Ms. Yashika J. Chawla

i

COMPANY PROFILE

Company Name : Engeniuspark Technologies Pvt. Ltd. (ESTPL)

Introduction to the Company

Engeniuspark Technologies Pvt. Ltd. (ESTPL) is an eminent IT firm located in Nashik, Maharasthra, India. The company was incorporated in 2014 and offers unique and affordable technology solutions to its customers. The firm specializes in Software development, Web application development, Mobile applications, Artificial Intelligence (AI), Internet of Things (IoT), Digital Marketing, and Smart technology solutions.

The firm is committed to adopting innovations and ensuring client satisfaction. Thus far, ESTPL has successfully delivered quality services to clients across national and international borders.

Company Overview

Particular Details

Company Name Engeniuspark Technologies Pvt. Ltd. (ESTPL)

Industry Information Technology and Software Services

Founded 2014

Headquarters Nashik, Maharashtra, India

Company Type Private Limited Company

Website www.estplnow.com

Core Domains Software Development, AI, IoT, Mobile Applications, Digital Marketing

Presence India, USA, UAE

Client Base National and International Clients

ii

History and Background

Vision

To become a globally recognized technology organization by delivering innovative, reliable, and sustainable digital solutions that empower businesses and contribute to technological advancement.

Mission

• To provide high-quality software and technology solutions.

• To leverage emerging technologies for business growth.

• To maintain excellence through innovation and continuous improvement.

• To build long-term relationships with clients through trust and professionalism.

• To deliver cost-effective and customer-centric services.

Core Values

• Innovation

• Integrity

Page 2 of 5

Figure 38: Plagiarism Scan Report (2 of 5)

• Customer Satisfaction
• Excellence
• Teamwork
• Continuous Learning
• Accountability
• Professionalism
Services Offered
• Software Development
• Web Development
iii
• Mobile Application Development
• Artificial Intelligence Solutions
• Internet of Things (IoT)
• Digital Marketing
• Aviation Solutions
• Testing and Quality Assurance
Organizational Structure
• Chief Executive Officer (CEO)
• Directors
• Project Managers
• Team Leaders
• Software Developers
• UI/UX Designers
• Quality Assurance Engineers
• Database Administrators
• Digital Marketing Team
• Technical Support Team
• Human Resource Department
Technologies Used
Frontend Technologies: HTML5, CSS3, JavaScript, React.js
Backend Technologies: Node.js, Express.js, Java, .NET Technologies
Database Technologies: MySQL, MongoDB
Development Tools: Visual Studio Code, Git, GitHub, Postman
iv
Emerging Technologies: Artificial Intelligence, Internet of Things (IoT), Cloud Com-
puting
Why Choose Engeniuspark Technologies Pvt. Ltd.
• Experienced and highly skilled professionals.
• Customer-centric development approach.
• Timely project delivery.
• Innovative and scalable solutions.
• Strong technical support.
• Global presence and industry expertise.
• Cost-effective services.
• Quality-driven development process.
Strengths of the Company
• Experienced technical professionals.
• Strong focus on innovation and research.
• Adoption of latest technologies.
• Excellent customer support.
• High-quality software development standards.
• Global client reach.
• Strong project management capabilities.
• Collaborative and learning-oriented work culture.

Figure 39: Plagiarism Scan Report (3 of 5)

Contact Information
v
Particular Details
Company Name Engeniuspark Technologies Pvt. Ltd.
Address Office No. 403, Shree Kalika, Opp. Kalika Mata Mandir, Above Haldiram & Zudio, Shingada Talav, Mumbai Naka, Renuka Nagar, Nashik, Maharash-tra – 422011
Contact Number +91 90117 55405
Website www.estplnow.com
Global Presence India, USA, UAE
Internship Exposure
In the course of the internship at Engeniuspark Technologies Pvt. Ltd., practical exposure was gained in software development methodologies, project management practices, web technologies, database management systems, and industrial development tools. The internship helped gain insights into actual software development environments and the professional work atmosphere.
Conclusion
Engeniuspark Technologies Pvt. Ltd. has been able to position itself as an emerging and innovative technology firm owing to the provision of top-notch digital services in diverse sectors. The company is committed to innovation, customer satisfaction, and technological superiority, which has enabled it to make significant contributions to the digital revolution.
vi
vii
viii
Abstract
In today's digital age, restaurants are increasingly adopting smart technologies to im-prove customer convenience and service efficiency. This project presents a Digital Menu Application that uses QR code technology along with a mobile-friendly Progressive Web Application (PWA). Instead of traditional printed menus, customers can simply scan a table-specific QR code to access the menu directly on their smartphones. This allows them to view item details such as descriptions, prices, nutritional information, and dietary categories without

Matched Source

Similarity 2%

Title:This is to certify that the project report entitled ..... being ...

This is to certify that the project report entitled ..... being submitted by Name1 Rollno1 of bonafied work carried out under my guidance and supervision. The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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Page 4 of 5

Figure 40: Plagiarism Scan Report (4 of 5)

Similarity 2%

Title:QR Code - Menu Maker AI - App Store - Apple

QR Code - Menu Maker AI - App Store - Apple Applehttps://apps.apple.com › app › qr-code-menu-maker-aiApplehttps://apps.apple.com › app › qr-code-menu-maker-ai

https://apps.apple.com/us/app/qr-code-menu-maker-ai/id6746864293

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https://www.scribd.com/document/722394198/Online-Art-Gallery

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https://www.scribd.com/document/799233872/Expanded-Ratio-Analysis-Project-Class12

Similarity 2%

Title:ZoomInfo

Ritas Law aims to build long-term relationships with clients through trust and professionalism.Explore more. FUN is OK's Social Media. Is this data correct ...

https://www.zoominfo.com/c/fun-is-ok/354968044

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Title:LinkedIn · Danveer Bhasin

South Africa has been able to position itself as an emerging outsourcing destination, not only for contact centers, but also high- end ...

https://www.linkedin.com/pulse/impact-sourcing-business-good-danveer-bhasin-1d

Check By: Dupli Checker

Page 5 of 5

Figure 41: Plagiarism Scan Report (5 of 5)

Appendix D

Team Information

Sr. No.	Name of Member	Email ID	Mobile Number	Role in the Project
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2	Sakshi Gambhire	—	—	Payment Gateway Integration and Payment Module Handling
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Table 5 : Team Information

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